

# SPATBench v33 — Supplementary Material

**Version:** v33 (2026-09-04 resync). Generated by scripts/synthesize\_v33\_supplementary.py from the machine-readable outputs in results/benchmark/v33/ and data/tcga/; do not edit numbers by hand — change the generating script and re-run.

**Contents:** Tables S1–S26, Figures S1–S12, Notes S1–S6.

## Table S1. Per-Fold Cross-Validation Setup

The v33 benchmark stores pooled out-of-fold (OOF) AUROC, 95% bootstrap CI, and AUPRC per method–cohort pair (results/benchmark/v33/benchmark\_v33.json). Per-fold AUROC values are fully determined by the archived split files (SHA1 recorded in results/run\_registry.json) and are reproducible with scripts/rerun\_v33.py. The fold structure per cohort:

| Cohort–endpoint                | n  | k (folds) | Endpoint note                                                       |
|--------------------------------|----|-----------|---------------------------------------------------------------------|
| Hugo 2016 RECIST (n=28)        | 28 | 5         | RECIST v1.1, 13 responders                                          |
| Lauss 2017 (ACT) RECIST (n=25) | 25 | 5         | RECIST v1.1, 10 responders                                          |
| Gide 2019 RECIST (n=73)        | 73 | 5         | RECIST v1.1, 40 responders                                          |
| Jung 2019 DCB (n=27)           | 27 | 5         | DCB, 6 responders (labels corrected v33: 1 = DCB)                   |
| Riaz 2017 cytolytic (n=43)     | 43 | 5         | cytolytic above median, 21/43                                       |
| Riaz 2017 RECIST (n=42)        | 42 | 2         | RECIST v1.1, 9 responders → k = 2 (primary); k = 2/3/5 in Table S16 |

## Table S2. Literature Comparison: Published vs. SPATBench Standardized AUROC (v33)

| Method          | Published AUROC | Published Cohort                 | SPATBench best (v33)        | Inflation | Reference                                 |
|-----------------|-----------------|----------------------------------|-----------------------------|-----------|-------------------------------------------|
| TIDE            | 0.80            | IMvigor210 (BLCA, n = 298)       | 0.752 (Gide 2019 RECIST)    | +0.05     | Jiang et al. 2018, Nature Medicine        |
| IMPRES          | 0.83            | Auslander 2018 (SKCM, n = 30–48) | 0.795 (Hugo 2016 RECIST)    | +0.04     | Auslander et al. 2018, Nature Medicine    |
| TMB (proxy)     | 0.64            | Various (n > 100)                | 0.500 constant proxy        | +0.14     | Goodman et al. 2017, Mol Cancer Ther      |
| PD-L1 (CD274)   | 0.65            | Multiple trials                  | 0.868 (Riaz 2017 cytolytic) | -0.22     | Topalian et al. 2012, NEJM                |
| GEP             | 0.67            | IMvigor210 (BLCA, n = 298)       | 0.968 (Riaz 2017 cytolytic) | -0.30     | Mariathasan et al. 2018, Nature           |
| ElasticNet (MI) | 0.80            | Multiple (n > 200)               | 0.968 (Riaz 2017 cytolytic) | -0.17     | Chowell et al. 2022, Nature Biotechnology |

*SPATBench best values are the maximum AUROC across the six v33 cohort–endpoints (leakage-free nested CV); the TMB proxy is a constant 0.5 baseline (real mutation data unavailable) and the PD-L1*

proxy is CD274 expression (real IHC scores unavailable).

**Caveat:** Published values come from different (typically larger) cohorts. This comparison illustrates cross-study variability, not a within-cohort evaluation. Inflation (published minus SPATBench best) ranges from -0.30 to +0.14.

**Table S3. Cohort Characteristics (Detailed, v33)**

| Attribute                      | Hugo 2016     | Riaz 2017                                                                              | Lauss 2017 (ACT) | Jung 2019                           | Gide 2019                             |
|--------------------------------|---------------|----------------------------------------------------------------------------------------|------------------|-------------------------------------|---------------------------------------|
| Accession                      | GSE78220      | GSE91061                                                                               | GSE100797        | GSE135222                           | mel_iatlas_gide_2019                  |
| Cancer                         | Melanoma      | Melanoma                                                                               | Melanoma         | NSCLC                               | Melanoma                              |
| n (pre-treatment)              | 28            | 43 (42 with RECIST)                                                                    | 25               | 27                                  | 73                                    |
| Treatment                      | Pembrolizumab | Nivolumab                                                                              | Adoptive TIL     | Anti-PD-1/PD-L1                     | Anti-PD-1 ± anti-CTLA-4               |
| Endpoint(s)                    | RECIST v1.1   | RECIST + cytolytic (same patients)                                                     | RECIST v1.1      | DCB vs NDB (1 = DCB, corrected v33) | RECIST v1.1                           |
| Responders                     | 13 (46%)      | 9 RECIST (of 42) / 21 cytolytic (of 43)                                                | 10 (40%)         | 6 DCB (22%)                         | 40 (55%)                              |
| Gene IDs (v33)                 | HGNC symbols  | Entrez → HGNC-mapped (21,769/22,187 genes)                                             | HGNC symbols     | Ensembl → HGNC-mapped               | HGNC symbols                          |
| CDS (declared genes GZMA/PRF1) | 0.449         | 0.685 RECIST / 0.954 cytolytic (archived); 0.409 RECIST on the HGNC matrix (Table S4b) | 0.781            | —                                   | 0.852 (first stored value; Table S4b) |
| k-fold (primary)               | 5             | 2 (RECIST) / 5 (cytolytic)                                                             | 5                | 5                                   | 5                                     |

**Table S4. Circularity Detection Score (CDS)**

**(a) Values as reported in manuscript Table 3 (archived computation)**

| Endpoint                | CDS   | C1 (MI) | C2 (ρ) | Risk (recalibrated, HIGH > 0.80) | Inflated? |
|-------------------------|-------|---------|--------|----------------------------------|-----------|
| Riaz 2017 cytolytic     | 0.954 | 0.573   | 0.845  | HIGH                             | Yes       |
| Lauss 2017 (ACT) RECIST | 0.781 | 0.049   | 0.294  | MODERATE                         | No        |
| Riaz 2017 RECIST        | 0.685 | 0.034   | 0.273  | MODERATE                         | No        |
| Hugo 2016 RECIST        | 0.449 | <0.001  | 0.164  | MODERATE                         | No        |

*Archived-provenance note: the two Riaz values were computed on the pre-v33 Entrez-ID matrices (endpoint genes recorded as Entrez 3001/5551); all other cohorts use HGNC symbol matrices. The recalibrated risk labels use the conservative HIGH > 0.80 boundary (candidate for CDS v2.0); under the shipped v1.1.0 threshold (HIGH > 0.70) Lauss (0.781) is labelled HIGH by the released tool.*

**(b) v33 recomputation with cds v1.1.0 on the HGNC matrices (uniform convention; scripts/supplementary\_v33\_experiments.py)**

All five evaluable endpoints, declared endpoint genes GZMA/PRF1, cds v1.1.0 (weights 30/30/40, shipped thresholds HIGH > 0.70, MODERATE 0.30–0.70), each with its cohort-matched random-label null (10 replicates, cds\_nulls\_v33.json; the manuscript Table 3 now reports the upgraded 200-replicate nulls from cds\_nulls\_200\_v33.json — all Table 3 readings are unchanged under the 200-replicate nulls):

| Endpoint (v33 matrix)   | CDS   | C1 (MI) | C2 (ρ) | C3    | null mean | null max | CDS vs null        | Risk (shipped v1.1.0) |
|-------------------------|-------|---------|--------|-------|-----------|----------|--------------------|-----------------------|
| Hugo 2016 RECIST        | 0.449 | 0.000   | 0.164  | 1.000 | 0.608     | 0.764    | at/below null mean | MODERATE              |
| Lauss 2017 (ACT) RECIST | 0.781 | 0.049   | 0.294  | 1.000 | 0.557     | 0.832    | above null mean    | HIGH                  |
| Gide 2019 RECIST        | 0.852 | 0.141   | 0.508  | 1.000 | 0.518     | 0.726    | far above null max | HIGH                  |
| Riaz 2017 RECIST (v33)  | 0.409 | 0.000   | 0.029  | 1.000 | 0.556     | 0.742    | at/below null mean | MODERATE              |
| Riaz 2017 cytolytic     | 0.954 | 0.573   | 0.845  | 1.000 | 0.510     | 0.764    | far above null max | HIGH                  |

**Recomputation notes (disclosed, not silently reconciled):**

1. Hugo (0.449), Lauss (0.781) and Riaz cytolytic (0.954) reproduce the archived scores exactly. Riaz RECIST recomputes to **0.409** on the HGNC matrix versus the archived 0.685 (Entrez space) — the archived value should be read as Entrez-space; the HGNC-space value is the one consistent with the v33 benchmark feature space.
2. **Gide 2019 RECIST has a stored CDS value for the first time: 0.852 (HIGH under both the shipped 0.70 and the conservative 0.80 boundary).** The manuscript narrative that CDS predicted Gide 'LOW' is not supported by this recomputation. Mechanistically, C1 and C2 read the strong association between the declared genes (GZMA/PRF1) and the clinical RECIST label in this large melanoma cohort — the same association that underlies the genuine GEP result (0.830,  $q < 0.001$ ) — as endpoint-gene dependency. This is the CDS limitation discussed in the manuscript (inability to separate genuine immune-biology signal from circularity) made concrete; the CDS risk vocabulary should be read as advisory (Methods), and the Gide CDS value is reported here for transparency pending the CDS v2.0 redesign.
3. Risk labels in (a) use the recalibrated HIGH > 0.80 boundary; the tool's shipped labels (column 'Risk (shipped v1.1.0)' above) flag Lauss and Gide as HIGH. The validated claim therefore needs the 0.80 boundary and, for Gide, does not hold under any current threshold — see note 2.

**(c) Synthetic null distribution (30 random-label replicates, quoted from Methods)**

Mean null CDS = 0.50 (SD = 0.09), 95th percentile = 0.65, 99th percentile = 0.71;  $\approx 3\%$  of random endpoints flagged HIGH under the shipped threshold due to C3 saturation at 1.0. The 30 null replicates were generated after the real endpoints had been inspected (retrospective calibration; Methods).

**(d) Synthetic five-scenario ranking (cds\_synthetic\_calibration\_v3.json)**

| Scenario | Circularity level                    | CDS   | Mean AUROC | Risk (shipped) |
|----------|--------------------------------------|-------|------------|----------------|
| 1        | Pure random endpoint                 | 0.697 | 0.468      | MODERATE       |
| 2        | $\sim 20\%$ circular, 80% biological | 0.820 | 0.691      | HIGH           |
| 3        | $\sim 50\%$ circular, 50% biological | 0.830 | 0.838      | HIGH           |
| 4        | $\sim 80\%$ circular, 20% biological | 0.832 | 0.788      | HIGH           |
| 5        | 100% circular endpoint               | 0.845 | 0.845      | HIGH           |

CDS ranks all five scenarios monotonically ( $0.697 < 0.820 < 0.830 < 0.832 < 0.845$ ). Endpoints with CDS  $\geq 0.80$  show inflated AUROC (0.691–0.845); the pure-random scenario sits at CDS 0.697 — above the manuscript's earlier '0.51' description, which came from the alternative v2 calibration (endpoint-weight parameterization, `cds_synthetic_calibration_v2.json`, scenario 1: CDS 0.513). The two calibrations parameterize circularity differently and are not interchangeable.

**Threshold disclosure:** one threshold set is shipped everywhere (HIGH  $> 0.70$ , MODERATE 0.30–0.70, LOW  $\leq 0.30$ ); the conservative 0.80 boundary (0/30 false positives on the synthetic null) is the candidate for CDS v2.0. Continuous CDS values should be reported; categorical labels are advisory.

**Table S5. Method Concordance — Cohen's  $\kappa$  (Hugo 2016, n = 28, v33)**

OOF predictions under the v33 leakage-free protocol (ElasticNet-MI frozen at  $C = 1.0$ , `l1_ratio = 0.1`); each method's continuous score binarized at its own median. All 10 method pairs are reported.

| Method pair                             | Agreement | Cohen's $\kappa$ |
|-----------------------------------------|-----------|------------------|
| IMPRES vs. GEP                          | 57%       | 0.143            |
| IMPRES vs. TIDE                         | 50%       | 0.000            |
| IMPRES vs. PD-L1 (CD274 expr.)          | 71%       | 0.429            |
| IMPRES vs. ElasticNet (MI)              | 57%       | 0.143            |
| GEP vs. TIDE                            | 50%       | 0.000            |
| GEP vs. PD-L1 (CD274 expr.)             | 79%       | 0.571            |
| GEP vs. ElasticNet (MI)                 | 57%       | 0.143            |
| TIDE vs. PD-L1 (CD274 expr.)            | 50%       | 0.000            |
| TIDE vs. ElasticNet (MI)                | 64%       | 0.286            |
| PD-L1 (CD274 expr.) vs. ElasticNet (MI) | 57%       | 0.143            |

8 of 10 pairs fall below the conventional fair-agreement threshold ( $\kappa = 0.40$ ); the highest is GEP vs. PD-L1 (CD274 expr.) ( $\kappa = 0.571$ ) — the concordance of two fixed scorers, not a trained model; the highest pair involving the trainable method is TIDE vs. ElasticNet (MI) ( $\kappa = 0.286$ ). At  $n = 28$  different methods flag largely different patients as responders.

**Table S6. IMPRES Leave-One-Pair-Out Analysis (Hugo 2016, n = 28, canonical 15-pair implementation, v33)**

Full IMPRES (15 pairs) AUROC = **0.795** (equals the v33 benchmark value; the score is deterministic). AUROC when each pair is dropped from the score:

| Dropped pair   | AUROC without pair | $\Delta$ from full | Direction         |
|----------------|--------------------|--------------------|-------------------|
| CD86-CD200     | 0.818              | +0.023             | Removing improves |
| CTLA4-TNFSF4   | 0.803              | +0.008             | Marginal          |
| CD274-C10orf54 | 0.795              | +0.000             | Marginal          |
| CD28-CD276     | 0.795              | +0.000             | Marginal          |
| CD40-CD28      | 0.795              | +0.000             | Marginal          |

| Dropped pair  | AUROC without pair | $\Delta$ from full | Direction      |
|---------------|--------------------|--------------------|----------------|
| TNFRSF14-CD86 | 0.795              | +0.000             | Marginal       |
| CD40-CD80     | 0.795              | +0.000             | Marginal       |
| CD80-TNFSF9   | 0.795              | +0.000             | Marginal       |
| CD40-CD274    | 0.790              | -0.005             | Marginal       |
| CD86-TNFSF4   | 0.785              | -0.010             | Marginal       |
| CD86-HAVCR2   | 0.779              | -0.015             | Removing hurts |
| CD27-PDCD1    | 0.777              | -0.018             | Removing hurts |
| CD28-CD86     | 0.769              | -0.026             | Removing hurts |
| PDCD1-TNFSF4  | 0.756              | -0.038             | Removing hurts |
| CD40-PDCD1    | 0.751              | -0.044             | Removing hurts |

All 15 canonical relations are retained in the shipped implementation. Simplifying the pair set on this cohort would constitute outcome-guided selection (data leakage) and is not performed. Interpretative caveat: Carter et al. (2019) reported potential training-set bias in the original IMPRES feature selection [30].

**Table S7. Expression-Scale Sensitivity (Hugo 2016, ElasticNet-MI, v33 protocol)**

| Scale                                                                   | AUROC (pooled OOF) | Source             |
|-------------------------------------------------------------------------|--------------------|--------------------|
| $\log_2(\text{FPKM} + 1)$ (default)                                     | 0.631              | benchmark_v33.json |
| Per-sample quantile normalization of the $\log_2(\text{FPKM}+1)$ matrix | 0.528              | recomputed (v33)   |

*The archived v28 table also reported a TPM row. TPM cannot be derived from the archived  $\log_2(\text{FPKM}+1)$  matrix (no raw counts or gene-length metadata are stored in the repository), so that row is omitted rather than reproduced from the superseded v28 pipeline.*

**Table S8. Cross-Validation Sensitivity (Hugo 2016, ElasticNet-MI, v33 protocol)**

| Strategy        | AUROC (pooled OOF) | Notes                           |
|-----------------|--------------------|---------------------------------|
| k = 3           | 0.564              | larger training folds           |
| k = 5 (default) | 0.631              | benchmark value (same protocol) |
| k = 10          | 0.549              | smaller training folds          |
| LOO (n = 28)    | 0.487              | extreme per-fold variance       |

All values fall within the bootstrap CI of the k = 5 estimate [0.417–0.857]; the ranking is not stable across strategies at n = 28, consistent with the seed-stability analysis (Table S12). Frozen hyperparameters; stratified splits with seed 42.

**Table S9. Full Benchmark Results, All Six Cohort–Endpoints (v33)**

Leakage-free nested CV (per-fold feature selection, inner 3-fold tuning), identical protocol for all cohorts. 95% bootstrap CIs (1,000 iterations).

| Cohort-endpoint                | Method           | AUROC | 95% CI        | AUPRC |
|--------------------------------|------------------|-------|---------------|-------|
| Hugo 2016 RECIST (n=28)        | IMPRES           | 0.795 | [0.597-0.947] | 0.709 |
| Hugo 2016 RECIST (n=28)        | GEP              | 0.446 | [0.228-0.672] | 0.476 |
| Hugo 2016 RECIST (n=28)        | TIDE             | 0.426 | [0.184-0.663] | 0.495 |
| Hugo 2016 RECIST (n=28)        | PD-L1 (CD274)    | 0.523 | [0.302-0.749] | 0.463 |
| Hugo 2016 RECIST (n=28)        | ElasticNet (MI)  | 0.631 | [0.417-0.857] | 0.624 |
| Hugo 2016 RECIST (n=28)        | ElasticNet (Var) | 0.528 | [0.318-0.761] | 0.492 |
| Lauss 2017 (ACT) RECIST (n=25) | IMPRES           | 0.610 | [0.387-0.826] | 0.460 |
| Lauss 2017 (ACT) RECIST (n=25) | GEP              | 0.693 | [0.455-0.912] | 0.593 |
| Lauss 2017 (ACT) RECIST (n=25) | TIDE             | 0.500 | [0.274-0.743] | 0.410 |
| Lauss 2017 (ACT) RECIST (n=25) | PD-L1 (CD274)    | 0.780 | [0.571-0.955] | 0.697 |
| Lauss 2017 (ACT) RECIST (n=25) | ElasticNet (MI)  | 0.700 | [0.486-0.896] | 0.576 |
| Lauss 2017 (ACT) RECIST (n=25) | ElasticNet (Var) | 0.627 | [0.373-0.840] | 0.553 |
| Gide 2019 RECIST (n=73)        | IMPRES           | 0.626 | [0.509-0.746] | 0.661 |
| Gide 2019 RECIST (n=73)        | GEP              | 0.830 | [0.734-0.915] | 0.847 |
| Gide 2019 RECIST (n=73)        | TIDE             | 0.752 | [0.639-0.859] | 0.785 |
| Gide 2019 RECIST (n=73)        | PD-L1 (CD274)    | 0.791 | [0.686-0.881] | 0.846 |
| Gide 2019 RECIST (n=73)        | ElasticNet (MI)  | 0.629 | [0.500-0.746] | 0.672 |
| Gide 2019 RECIST (n=73)        | ElasticNet (Var) | 0.605 | [0.478-0.731] | 0.625 |
| Jung 2019 DCB (n=27)           | IMPRES           | 0.583 | [0.380-0.750] | 0.256 |
| Jung 2019 DCB (n=27)           | GEP              | 0.786 | [0.573-0.960] | 0.439 |
| Jung 2019 DCB (n=27)           | TIDE             | 0.278 | [0.083-0.527] | 0.176 |
| Jung 2019 DCB (n=27)           | PD-L1 (CD274)    | 0.659 | [0.464-0.861] | 0.300 |
| Jung 2019 DCB (n=27)           | ElasticNet (MI)  | 0.476 | [0.152-0.792] | 0.340 |
| Jung 2019 DCB (n=27)           | ElasticNet (Var) | 0.476 | [0.100-0.818] | 0.295 |
| Riaz 2017 cytolytic (n=43)     | IMPRES           | 0.660 | [0.493-0.811] | 0.649 |
| Riaz 2017 cytolytic (n=43)     | GEP              | 0.968 | [0.915-1.000] | 0.971 |
| Riaz 2017 cytolytic (n=43)     | TIDE             | 0.470 | [0.282-0.653] | 0.539 |
| Riaz 2017 cytolytic (n=43)     | PD-L1 (CD274)    | 0.868 | [0.749-0.958] | 0.890 |
| Riaz 2017 cytolytic (n=43)     | ElasticNet (MI)  | 0.968 | [0.909-1.000] | 0.975 |
| Riaz 2017 cytolytic (n=43)     | ElasticNet (Var) | 0.985 | [0.954-1.000] | 0.985 |
| Riaz 2017 RECIST (n=42)        | IMPRES           | 0.613 | [0.381-0.835] | 0.308 |
| Riaz 2017 RECIST (n=42)        | GEP              | 0.503 | [0.305-0.710] | 0.256 |
| Riaz 2017 RECIST (n=42)        | TIDE             | 0.466 | [0.222-0.723] | 0.295 |
| Riaz 2017 RECIST (n=42)        | PD-L1 (CD274)    | 0.545 | [0.330-0.759] | 0.259 |
| Riaz 2017 RECIST (n=42)        | ElasticNet (MI)  | 0.569 | [0.370-0.755] | 0.255 |
| Riaz 2017 RECIST (n=42)        | ElasticNet (Var) | 0.465 | [0.208-0.704] | 0.317 |

The v33 benchmark JSON stores AUROC/CI/AUPRC per cell. The additional per-cell metrics of the archived v28 table (F1, sensitivity, specificity, Brier) are not part of the v33 JSON schema; representative v33 clinical-utility metrics for ElasticNet-Var on Hugo 2016 are below.

**Clinical utility, ElasticNet-Var on Hugo 2016 (v33 recomputed, hugo\_envar\_clinical\_v33.json):**

| Metric                                                       | Value (v33)        | Archived v28 value |
|--------------------------------------------------------------|--------------------|--------------------|
| Sensitivity / Specificity (0.5 threshold)                    | 0.538 / 0.533      | 0.385 / 0.600      |
| Brier score                                                  | 0.337              | 0.339              |
| DCA net benefit @ p <sub>t</sub> = 0.10 (model vs treat-all) | 0.345 vs 0.405     | 0.155 vs 0.405     |
| DCA net benefit @ p <sub>t</sub> = 0.20 (model vs treat-all) | 0.259 vs 0.330     | 0.125 vs —         |
| DCA net benefit @ p <sub>t</sub> = 0.50 (model vs treat-all) | 0.000 vs -0.071    | -0.036 vs —        |
| NRI vs. random baseline (total)                              | -0.113 (p = 0.658) | +0.985             |
| IDI vs. random baseline (total)                              | -0.029             | -0.015             |

*Under the v33 protocol the model's net benefit remains positive but below 'treat all' at 10–20% thresholds; NRI vs. random is negative and non-significant (p = 0.658). The archived +0.985 NRI was a v28-pipeline artifact and is superseded.*

**Clinical utility, IMPRES on Hugo 2016 (the cohort's only FDR-significant method; hugo\_impres\_clinical\_v33.json, same protocol and conventions):**

| Metric                                                        | Value (v33)        |
|---------------------------------------------------------------|--------------------|
| Sensitivity / Specificity (0.5 threshold)                     | 0.923 / 0.267      |
| Brier score                                                   | 0.231              |
| DCA net benefit at p <sub>t</sub> = 0.10 (model vs treat-all) | 0.405 vs 0.405     |
| DCA net benefit at p <sub>t</sub> = 0.20 (model vs treat-all) | 0.330 vs 0.330     |
| DCA net benefit at p <sub>t</sub> = 0.50 (model vs treat-all) | 0.036 vs -0.071    |
| NRI vs. random baseline (total)                               | -0.200 (p = 0.900) |
| IDI vs. random baseline (total)                               | -0.073             |

*The ranking advantage of the cohort's only FDR-significant method (AUROC 0.795) does not translate into decision utility: at the 0.5 threshold it labels 92% of patients responders, its net benefit coincides with treat-all at 10–20% thresholds, and its reclassification metrics do not exceed the random baseline. The IMPRES score is an uncalibrated fraction, so these metrics reflect threshold behavior rather than calibrated risk; the same reading follows either way — a high AUROC alone does not establish clinical usefulness.*

**Survival stratification (Hugo 2016):**

| Quantity             | True OS (iAtlas OS_STATUS) | v33 approximation (0 < OS ≤ 400 d) |
|----------------------|----------------------------|------------------------------------|
| n / events           | 26 / 12                    | 25 / 11                            |
| IMPRES KM log-rank p | 0.554                      | 0.054                              |
| GEP KM log-rank p    | 0.475                      | 0.428                              |
| RECIST KM log-rank p | 0.0003                     | 0.0003                             |
| IMPRES Cox HR per SD | 0.631 [0.325–1.227]        | 0.598 [0.302–1.186]                |
| GEP Cox HR per SD    | 1.03 [0.562–1.887]         | 1.208 [0.686–2.127]                |

*True-event column: n = 26, 12 deaths, events from the iAtlas harmonization of Hugo 2016 (hugo\_survival\_true.json; 24/27 local OS\_days agree with iAtlas OS\_MONTHS (±60 d); matching greedy one-to-one expression-profile correlation; min margin -0.033). The approximation column reproduces the v33 convention and is retained for continuity; conclusions are unchanged (IMPRES does not significantly stratify OS on either convention).*

**Table S10. Immune Deconvolution Signature Marker Genes (43 Genes, 14 Signatures)**

*Provenance note (2026-09-18): the archived header of this table read “64 Genes, 14 Signatures”; the correct count of unique marker genes in the table is 43, and the header has been corrected accordingly.*

| Signature       | Marker genes                      |
|-----------------|-----------------------------------|
| CD8 T cells     | CD8A, CD8B                        |
| CD4 T cells     | CD4                               |
| B cells         | CD19, CD79A, MS4A1                |
| NK cells        | NKG7, KLRD1, KLRF1                |
| M1 Macrophages  | NOS2, IL12A, TNF                  |
| M2 Macrophages  | CD163, MRC1, MSR1                 |
| Tregs           | FOXP3, IL2RA                      |
| Dendritic cells | CD1C, CLEC10A, ITGAX              |
| Neutrophils     | ELANE, MPO, CEACAM8               |
| Cytolytic score | GZMA, PRF1, GNLY                  |
| IFNG response   | IFNG, CXCL10, CXCL9, IDO1, STAT1  |
| TCR signaling   | CD3D, CD3E, CD3G, TRAC            |
| Checkpoint      | PDCD1, CTLA4, LAG3, TIGIT, HAVCR2 |
| TLS             | CXCL13, CCL19, CCL21              |

Signatures are the mean  $\log_2(\text{FPKM} + 1)$  expression of the marker genes found in each cohort (no training, continuous score).

**Signature AUROCs across all six cohort–endpoints (v33; HGNC-mapped matrices for Riaz and Jung)**

| Signature         | Hugo '16<br>RECIST (n=28) | Lauss '17 (ACT)<br>RECIST (n=25) | Gide '19<br>RECIST (n=73) | Jung '19<br>DCB (n=27) | Riaz '17<br>cytolytic (n=43) | Riaz '17<br>RECIST (n=42) |
|-------------------|---------------------------|----------------------------------|---------------------------|------------------------|------------------------------|---------------------------|
| CD8 T cells       | 0.492                     | 0.707                            | 0.831                     | 0.746                  | 0.987                        | 0.465                     |
| CD4 T cells       | 0.467                     | 0.693                            | 0.647                     | —                      | 0.814                        | 0.481                     |
| B cells           | 0.549                     | 0.627                            | 0.701                     | 0.532                  | 0.859                        | 0.512                     |
| NK cells          | 0.513                     | 0.613                            | 0.780                     | 0.698                  | 0.998                        | 0.458                     |
| M1<br>Macrophages | 0.544                     | 0.800                            | 0.564                     | —                      | 0.675                        | 0.478                     |
| M2<br>Macrophages | 0.308                     | 0.660                            | 0.660                     | —                      | 0.788                        | 0.519                     |
| Tregs             | 0.385                     | 0.727                            | 0.692                     | —                      | 0.900                        | 0.562                     |
| Dendritic cells   | 0.600                     | 0.667                            | 0.665                     | —                      | 0.864                        | 0.481                     |
| Neutrophils       | 0.426                     | 0.503                            | 0.557                     | —                      | 0.545                        | 0.429                     |
| Cytolytic score   | 0.415                     | 0.653                            | 0.764                     | 0.738                  | 1.000                        | 0.475                     |
| IFNG response     | 0.431                     | 0.707                            | 0.834                     | 0.603                  | 0.939                        | 0.522                     |
| TCR signaling     | 0.554                     | 0.687                            | 0.765                     | —                      | 0.987                        | 0.475                     |

| Signature  | Hugo '16<br>RECIST (n=28) | Lauss '17 (ACT)<br>RECIST (n=25) | Gide '19<br>RECIST (n=73) | Jung '19<br>DCB (n=27) | Riaz '17<br>cytolytic (n=43) | Riaz '17<br>RECIST (n=42) |
|------------|---------------------------|----------------------------------|---------------------------|------------------------|------------------------------|---------------------------|
| Checkpoint | 0.441                     | 0.700                            | 0.784                     | 0.706                  | 0.968                        | 0.468                     |
| TLS        | 0.513                     | 0.707                            | 0.686                     | —                      | 0.855                        | 0.468                     |

*Caveats: the Riaz cytolytic column is scored against the cytolytic-above-median endpoint, so the Cytolytic-score cell there is circular by construction ( $CDS = 0.954$ ). Genes absent from a cohort are skipped and the mean is over the found genes; per-signature found/total counts are in `supplementary_v33_recompute.json`. The archived statement that immune signatures 'could not be computed on Riaz (Entrez) or Jung (Ensembl)' is superseded: the v33 HGNC-mapped matrices make all 14 signatures computable on every cohort.*

### Table S11. Complete Permutation Test Matrix (v33)

Fixed scorers (IMPRES, GEP, TIDE, PD-L1): 50,000 label shuffles against the fixed score (mathematically equivalent to the full-pipeline null because the score is label-independent; the equivalence was verified on Hugo IMPRES). Trainable methods (ElasticNet-MI, ElasticNet-Var): full-pipeline label shuffles with frozen hyperparameters, per-cell n. BH correction within each cohort.

| Method | Cohort | n | Obs AUROC | Perm-obs AUROC | p | n (shuffles) | BH q | Significant |

*Trainable-method p-values are computed against the frozen-hyperparameter Perm-obs AUROC (right column), not the tuned benchmark Obs AUROC; see the footnote below.*

|                 |                                |    |       |       |         |       |       |     |
|-----------------|--------------------------------|----|-------|-------|---------|-------|-------|-----|
| IMPRES          | Hugo 2016 RECIST (n=28)        | 28 | 0.795 | 0.795 | 0.0029  | 50000 | 0.017 | Yes |
| IMPRES          | Lauss 2017 (ACT) RECIST (n=25) | 25 | 0.610 | 0.610 | 0.1893  | 50000 | 0.227 |     |
| IMPRES          | Gide 2019 RECIST (n=73)        | 73 | 0.626 | 0.626 | 0.03168 | 50000 | 0.032 | Yes |
| IMPRES          | Jung 2019 DCB (n=27)           | 27 | 0.583 | 0.583 | 0.4046  | 50000 | 0.486 |     |
| IMPRES          | Riaz 2017 cytolytic (n=43)     | 43 | 0.660 | 0.660 | 0.03224 | 50000 | 0.039 | Yes |
| IMPRES          | Riaz 2017 RECIST (n=42)        | 42 | 0.613 | 0.613 | 0.1535  | 50000 | 0.453 |     |
| GEP             | Hugo 2016 RECIST (n=28)        | 28 | 0.446 | 0.446 | 0.6916  | 50000 | 0.755 |     |
| GEP             | Lauss 2017 (ACT) RECIST (n=25) | 25 | 0.693 | 0.693 | 0.0567  | 50000 | 0.085 |     |
| GEP             | Gide 2019 RECIST (n=73)        | 73 | 0.830 | 0.830 | 2e-05   | 50000 | 0.000 | Yes |
| GEP             | Jung 2019 DCB (n=27)           | 27 | 0.786 | 0.786 | 0.01692 | 50000 | 0.102 |     |
| GEP             | Riaz 2017 cytolytic (n=43)     | 43 | 0.968 | 0.968 | 2e-05   | 50000 | 0.000 | Yes |
| GEP             | Riaz 2017 RECIST (n=42)        | 42 | 0.503 | 0.503 | 0.4889  | 50000 | 0.587 |     |
| TIDE            | Hugo 2016 RECIST (n=28)        | 28 | 0.426 | 0.426 | 0.7552  | 50000 | 0.755 |     |
| TIDE            | Lauss 2017 (ACT) RECIST (n=25) | 25 | 0.500 | 0.500 | 0.5103  | 50000 | 0.510 |     |
| TIDE            | Gide 2019 RECIST (n=73)        | 73 | 0.752 | 0.752 | 0.00014 | 50000 | 0.000 | Yes |
| TIDE            | Jung 2019 DCB (n=27)           | 27 | 0.278 | 0.278 | 0.9521  | 50000 | 0.952 |     |
| TIDE            | Riaz 2017 cytolytic (n=43)     | 43 | 0.470 | 0.470 | 0.6365  | 50000 | 0.637 |     |
| TIDE            | Riaz 2017 RECIST (n=42)        | 42 | 0.466 | 0.466 | 0.6223  | 50000 | 0.622 |     |
| PD-L1 (CD274)   | Hugo 2016 RECIST (n=28)        | 28 | 0.523 | 0.523 | 0.428   | 50000 | 0.642 |     |
| PD-L1 (CD274)   | Lauss 2017 (ACT) RECIST (n=25) | 25 | 0.780 | 0.780 | 0.0096  | 50000 | 0.058 |     |
| PD-L1 (CD274)   | Gide 2019 RECIST (n=73)        | 73 | 0.791 | 0.791 | 2e-05   | 50000 | 0.000 | Yes |
| PD-L1 (CD274)   | Jung 2019 DCB (n=27)           | 27 | 0.659 | 0.659 | 0.1299  | 50000 | 0.390 |     |
| PD-L1 (CD274)   | Riaz 2017 cytolytic (n=43)     | 43 | 0.868 | 0.868 | 2e-05   | 50000 | 0.000 | Yes |
| PD-L1 (CD274)   | Riaz 2017 RECIST (n=42)        | 42 | 0.545 | 0.545 | 0.3463  | 50000 | 0.519 |     |
| ElasticNet (MI) | Hugo 2016 RECIST (n=28)        | 28 | 0.631 | 0.528 | 0.2853  | 5000  | 0.642 |     |
| ElasticNet (MI) | Lauss 2017 (ACT) RECIST (n=25) | 25 | 0.700 | 0.613 | 0.03579 | 5000  |       |     |

| 0.072 | | ElasticNet (MI) | Gide 2019 RECIST (n=73) | 73 | 0.629 | 0.662 | 0.0232 | 5000 | 0.028 | **Yes** | | ElasticNet (MI) | Jung 2019 DCB (n=27) | 27 | 0.476 | 0.564 | 0.2364 | 2000 | 0.460 | | ElasticNet (MI) | Riaz 2017 cytolytic (n=43) | 43 | 0.968 | 0.933 | 0.001996 | 500 | 0.003 | **Yes** | | ElasticNet (MI) | Riaz 2017 RECIST (n=42) | 42 | 0.569 | 0.616 | 0.0976 | 2000 | 0.453 | | ElasticNet (Var) | Hugo 2016 RECIST (n=28) | 28 | 0.528 | 0.497 | 0.3635 | 5000 | 0.642 | | ElasticNet (Var) | Lauss 2017 (ACT) RECIST (n=25) | 25 | 0.627 | 0.627 | 0.03279 | 5000 | 0.072 | | ElasticNet (Var) | Gide 2019 RECIST (n=73) | 73 | 0.605 | 0.605 | 0.02 | 500 | 0.028 | **Yes** | | ElasticNet (Var) | Jung 2019 DCB (n=27) | 27 | 0.476 | 0.524 | 0.3064 | 2000 | 0.460 | | ElasticNet (Var) | Riaz 2017 cytolytic (n=43) | 43 | 0.985 | 0.974 | 0.000999 | 1000 | 0.002 | **Yes** | | ElasticNet (Var) | Riaz 2017 RECIST (n=42) | 42 | 0.465 | 0.545 | 0.2267 | 2000 | 0.453 | |

*All 36 of 36 method × cohort-endpoint cells carry a recorded permutation p (the previously n.c. trainable cells were completed by scripts/fill\_nc\_cells.py and merged by scripts/merge\_nc\_cells.py; seed 42; frozen hyperparameters). The Obs AUROC column shows the benchmark AUROC (tune-primary protocol). The Perm-obs AUROC column shows the observed AUROC under the permutation protocol. For the trainable cells re-run with frozen hyperparameters (recorded in results/benchmark/v33/nc\_cells/.json, summarized in perm\_obs\_auroc\_v33.json) it deviates from the benchmark value by 0.000–0.103 (largest: Hugo ElasticNet-MI 0.528 vs 0.631 — frozen-versus-tuned hyperparameters, the same deviation class as the Gide ElasticNet-MI cell, which re-run at 5,000 shuffles gives 0.662 vs the benchmark 0.629, i.e. +0.033). On 2026-09-18 all eight clinical trainable cells were extended to a uniform minimum of 2,000 full-pipeline shuffles (four cells at 5,000) as a pre-registered resolution-hardening check (scripts/extend\_nc\_cells\_5000.py, merged by scripts/merge\_nc\_cells\_ext.py): no cell changed its significance verdict, and the smallest within-cohort trainable q at  $n \leq 43$  tightened from 0.108 to 0.0716 (Lauss); the merge is reversible via the pre-merge snapshot permutation\_v33.pre\_ext\_backup.json. For the remaining four trainable cells (Gide MI/Var and Riaz cytolytic MI/Var) the permutation-run AUROC reproduces the benchmark value within 0.0003. For fixed scorers the observed score is the benchmark AUROC by construction (the score is label-independent); PD-L1 cells use the fixed-scorer protocol (scripts/pd11\_perm\_v33.py).\**

Cells surviving within-cohort BH ( $q < 0.05$ ): 12. Of these, the trainable-method cells are: ElasticNet (MI) on Gide 2019 RECIST (n=73) (0.629,  $q = 0.028$ ), ElasticNet (Var) on Gide 2019 RECIST (n=73) (0.605,  $q = 0.028$ ), ElasticNet (MI) on Riaz 2017 cytolytic (n=43) (0.968,  $q = 0.003$ ), ElasticNet (Var) on Riaz 2017 cytolytic (n=43) (0.985,  $q = 0.002$ ). No trainable method survives FDR at  $n \leq 43$ ; Gide ( $n = 73$ ) is the first sample size at which a training-based signature becomes detectable, consistent with the power analysis.

**Table S12. Random Seed Stability (10 stratified seeds, MI vs. variance, Hugo 2016, v33 protocol)**

| Metric                         | Value  |
|--------------------------------|--------|
| MI mean AUROC (10 seeds)       | 0.515  |
| MI SD across seeds             | 0.066  |
| Var mean AUROC (10 seeds)      | 0.523  |
| Var SD across seeds            | 0.038  |
| $\Delta$ AUROC mean (MI – Var) | -0.008 |
| $\Delta$ AUROC SD across seeds | 0.050  |

| Metric                | Value                                                                                |
|-----------------------|--------------------------------------------------------------------------------------|
| MI wins (of 10 seeds) | 4                                                                                    |
| Conclusion            | $\Delta$ AUROC is within cross-seed noise; rankings are seed-dependent at $n = 28$ . |

**Table S13. Learning Curve Analysis (Hugo 2016 subsampling, ElasticNet-Var, v33 protocol, 10 seeds per size)**

| n  | % of full | AUROC (mean over seeds) | SD across seeds | Mean cross-fold SD |
|----|-----------|-------------------------|-----------------|--------------------|
| 8  | 29%       | 0.512                   | 0.211           | 0.188              |
| 14 | 50%       | 0.487                   | 0.191           | 0.333              |
| 21 | 75%       | 0.524                   | 0.125           | 0.269              |
| 28 | 100%      | 0.523                   | 0.038           | 0.221              |

**Caveat:** at  $n = 8$  the fold count follows the  $k = \min(5, n_{\text{pos}}, n_{\text{neg}})$  convention ( $k = 4$  here); the small- $n$  point illustrates overfitting behavior (high variance) rather than a precise performance estimate. AUROC decreases while variance contracts as  $n$  grows — the expected small- $n$  pattern.

**Table S14. Effect-Size Comparison — Riaz 2017 (v33, HGNC-mapped)**

Effect-size comparison framework (manuscript Methods): ranges of AUROC attributable to each design source, not variance-component percentages.

| Source of variation                                           | Estimated AUROC effect                                             | Notes                                                                                                                                                           |
|---------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response definition (cytolytic $\rightarrow$ RECIST), largest | $\Delta$ AUROC = 0.520 (ElasticNet-Var: 0.985 $\rightarrow$ 0.465) | dominant; every method sharing endpoint genes collapses (GEP 0.968 $\rightarrow$ 0.503); IMPRES $\Delta = 0.047$ (no gene overlap with the endpoint definition) |
| Between-method, same RECIST endpoint (Riaz)                   | max $\Delta$ AUROC = 0.147 (IMPRES 0.613 vs TIDE 0.466)            | endpoint effect exceeds it by 3.5 $\times$                                                                                                                      |
| Between-method, same cytolytic endpoint (Riaz)                | max $\Delta$ AUROC = 0.515 (EN-Var vs TIDE)                        | on the circular endpoint (CDS 0.954)                                                                                                                            |
| Between-method, cross-cohort spread                           | up to 0.508 (Jung: GEP vs TIDE)                                    | why the endpoint effect must be compared within-cohort                                                                                                          |
| Feature selection (MI vs Var), Hugo 2016                      |                                                                    | $\Delta$ AUROC                                                                                                                                                  |

The v28 random-effects variance decomposition (68%/21%/11%) and the archived ' $\Delta = 0.545$  (EN-MI)' effect size are superseded; the v33 largest endpoint effect is 0.520 (EN-Var).

Replication-cohort effect ratios (GEP endpoint  $\Delta$  divided by the largest between-method gap on the same RECIST endpoint, all six methods): Liu 2019  $\approx 2.2\times$  (0.425/0.195) and IMvigor210  $\approx 2.0\times$  (0.354/0.180) — the direction of the Riaz 3.5 $\times$  ratio replicates, at a smaller magnitude (Table S19/S20).

**Table S15. TCGA Cox Regression — 12 Immune Signatures  $\times$  4 Cancer Types**

## (2,491 Patients, v33)

COAD has been re-acquired from the GDC current index (STAR counts, primary tumours, GENCODE v36; `scripts/build_coad_gdc.py`) after the original non-TCGA raw files forced the v33 withdrawal. SKCM/BRCA/LUAD use the unchanged cBioPortal RSEM matrices; COAD uses GDC STAR FPKM (within-cancer univariate Cox, disclosed source difference). SKCM uses the all-tumour sample policy (n = 441). BH correction within each cancer type (12 tests).

| Signature                           | SKCM<br>HR   | SKCM BH<br>q | BRCA<br>HR  | BRCA BH<br>q | LUAD<br>HR  | LUAD BH<br>q | COAD<br>HR  | COAD BH<br>q |
|-------------------------------------|--------------|--------------|-------------|--------------|-------------|--------------|-------------|--------------|
| M1 Macrophage                       | 0.795        | 0.0015       | 0.977       | 0.787        | 0.897       | 0.227        | 0.935       | 0.915        |
| IMPRES-like                         | 0.807        | 8.65e-08     | 0.901       | 0.112        | 0.896       | 0.227        | 1.036       | 0.915        |
| NK cells                            | 0.810        | 1.77e-07     | 0.889       | 0.0996       | 0.931       | 0.337        | 1.124       | 0.915        |
| Myeloid                             | 0.822        | 2.07e-05     | 0.932       | 0.507        | 0.899       | 0.227        | 0.986       | 0.915        |
| IFNG response                       | 0.827        | 1.24e-08     | 0.908       | 0.0922       | 0.971       | 0.564        | 0.965       | 0.915        |
| Treg                                | 0.827        | 5.1e-06      | 0.949       | 0.417        | 0.924       | 0.227        | 0.942       | 0.915        |
| Checkpoint                          | 0.829        | 8.65e-08     | 0.905       | 0.106        | 0.908       | 0.227        | 1.021       | 0.915        |
| Cytolytic                           | 0.863        | 3.01e-06     | 0.899       | 0.0722       | 0.946       | 0.366        | 0.970       | 0.915        |
| CD8 T cells                         | 0.867        | 2.56e-07     | 0.873       | 0.0205       | 0.916       | 0.227        | 1.177       | 0.915        |
| B cells                             | 0.904        | 5.46e-05     | 0.885       | 0.00383      | 0.873       | 0.00284      | 1.153       | 0.915        |
| TLS                                 | 0.928        | 0.00269      | 0.889       | 0.00383      | 0.941       | 0.227        | 1.040       | 0.915        |
| Stromal                             | 1.086        | 0.0903       | 1.026       | 0.76         | 1.089       | 0.227        | 1.030       | 0.915        |
| <b>Significant (BH q &lt; 0.05)</b> | <b>11/12</b> |              | <b>3/12</b> |              | <b>1/12</b> |              | <b>0/12</b> |              |

Cohort details: SKCM n = 441 (212 events); BRCA n = 1082 (151 events); LUAD n = 510 (185 events); COAD n = 458 (102 events). The stromal signature is the only non-significant one in SKCM and the only one with HR > 1 (1.086) — an internal negative control. HR < 1 = protective per 1 SD of the z-standardized signature. Univariate models; machine-readable in `data/tcga/tcga_cox_all.json`.

**Sample-type decomposition (SKCM).** The v33 SKCM policy is all-tumour samples, metastatic-dominant, while COAD (GDC) is primary-only — a composition difference the cross-cancer gradient partially conflates with cancer type. A three-arm decomposition of the identical 12-signature Cox + BH pipeline (`scripts/tcga_skcm_sampletype_sensitivity.py`; computed on cBioPortal-API-refetched matrices, the all-tumour arm reproducing the archived 11/12 with HR 0.795–0.928 exactly and thereby validating the refetch): all-tumour n = 434 (212 events), 11/12 BH-significant; **metastatic-only n = 360 (187 events), 11/12 BH-significant (HR 0.789–0.934)**; primary-only n = 76 (27 events), 0/12. The SKCM signal is carried by the metastatic arm and is internally robust to the composition question; the SKCM-versus-COAD sides of the gradient should be read as metastatic-dominant versus primary-only disease.

**Multivariate sensitivity (SKCM).** Cox models adjusted for age and AJCC stage (n = 390, 191 events; signature, age and stage z-standardized; stage parsed from AJCC codes to ordinal 1–4): **8/12 signatures remain BH-significant** with stronger protective HRs (0.569–0.704) than the univariate models — the immune-prognostic signal is not a stage- or age-proxy (`scripts/tcga_skcm_multivariate.py`, `SKCM_cox_multivariate.json`).

**Table S16. k-Fold Robustness — Riaz RECIST (ElasticNet-Var, 10 stratified seeds per k, v33)**

| k | Mean AUROC | SD    | n seeds |
|---|------------|-------|---------|
| 2 | 0.389      | 0.105 | 10      |
| 3 | 0.397      | 0.064 | 10      |
| 5 | 0.398      | 0.088 | 10      |

All values remain far below 0.60: the cytolytic → RECIST collapse is not an artifact of the  $k = 2$  constraint. The archived v28 k-fold JSON (0.51 / 0.318 / 0.361) is superseded by this regeneration (riaz\_recist\_kfold\_v33.json).

**Table S17. CDS Retrospective Assessment on Published Benchmark Studies**

**(a) Archived assessment (what CDS would have told each study before its benchmark)**

| Study                                                                                                                                                                                                                         | Cohort used            | Published AUROC | CDS (computable / estimated) | Was inflated? | CDS correct?       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------|------------------------------|---------------|--------------------|
| Auslander et al. 2018 (IMPRES)                                                                                                                                                                                                | Hugo 2016 (validation) | 0.83            | 0.449                        | False         | True               |
| Jiang et al. 2018 (TIDE)                                                                                                                                                                                                      | IMvigor210 (BLCA)      | 0.8             | 0.450                        | Unknown       | Unverified         |
| Mariathasan et al. 2018 (GEP)                                                                                                                                                                                                 | IMvigor210 (BLCA)      | 0.67            | 0.450                        | Unknown       | Unverified         |
| Chowell et al. 2022 (ElasticNet)                                                                                                                                                                                              | Multiple (n>200)       | 0.8             | 0.500                        | Partially     | Partially verified |
| Liang et al. 2026                                                                                                                                                                                                             | Multiple cohorts       | ~0.5-0.7        | —                            | Variable      | Unknown            |
| <i>Rows upgraded by the computed values in (b): the Jiang/Mariathasan IMvigor210 rows move from estimated to computed (RESPONSE CDS 0.434, reading unchanged); the Chowell and Liang multi-cohort rows remain unverified.</i> |                        |                 |                              |               |                    |

**(b) E3 update — computed CDS on cohorts accessible through the public iAtlas harmonizations (scripts/e3\_literature\_backtest.py)**

The archived assessment treated IMvigor210 and the Van Allen samples as inaccessible. They are accessible through cBioPortal (iAtlas), so the assessment is now upgraded from 'estimated' to computed. Expression values are cBioPortal z-scores — the CDS statistics (rank MI, Spearman) are invariant to per-gene affine transforms. Declared genes GZMA/PRF1; 10-replicate random-label null per cohort; cytolytic surrogate (median split of the declared-gene mean) as positive control.

| Cohort                   | Endpoint | n (responders) | CDS   | null mean | null max | Cytolytic control | Interpretation                                 |
|--------------------------|----------|----------------|-------|-----------|----------|-------------------|------------------------------------------------|
| IMvigor210 (blca_iatlas) | RESPONSE | 297 (68)       | 0.434 | 0.449     | 0.496    | 0.953             | at/below null mean                             |
| Liu 2019 (mel_iatlas)    | RESPONSE | 122 (48)       | 0.721 | 0.623     | 0.728    | 0.949             | above null mean                                |
| Van Allen 2015 samples   | —        | —              | —     | —         | —        | —                 | expression not accessible in the harmonization |

*In every cohort the cytolytic positive control sits far above its null, while the clinical endpoints sit at or within theirs — the internal finding (Table S4b) extends to independent studies. IMvigor210 and Liu 2019 are thereby upgraded from 'estimated' to computed CDS; the Van Allen samples remain expression-inaccessible in the harmonization (mutation-only), so the archived inaccessibility statement stands for that study; the endpoints used are the cohorts' published response annotations (IMvigor210 and Liu 2019: CR/PR vs SD/PD; RECIST responder vs non-responder).*

**Table S18. CDS Negative-Control Stress Test — 30 Real Gene-Program Pseudo-Endpoints per Cohort (E2)**

Design (manuscript Methods 'Negative-control stress test'): 30 canonical gene programs per cohort, dichotomised at the median of the program-mean expression (the same construction as the cytolytic surrogate), program genes declared as endpoint genes. Reference rows: the cytolytic surrogate positive control and a 10-replicate random-label null. Machine-readable: cds\_negative\_controls.json.

**(a) Summary**

| Cohort                  | Pseudo-endpoints HIGH | CDS range (min–max) | Mean  | Cytolytic control | Null mean | Null max |
|-------------------------|-----------------------|---------------------|-------|-------------------|-----------|----------|
| Gide 2019 RECIST (n=73) | 30/30                 | 0.792–0.942         | 0.861 | 0.955             | 0.518     | 0.726    |
| Hugo 2016 RECIST (n=28) | 30/30                 | 0.801–0.946         | 0.869 | 0.923             | 0.608     | 0.764    |

**(b) Per-endpoint results**

| Program                  | Family | Gide CDS | Gide risk | Hugo CDS | Hugo risk |
|--------------------------|--------|----------|-----------|----------|-----------|
| A1 fibrillar collagens   | A      | 0.942    | HIGH      | 0.946    | HIGH      |
| A2 fibroblast activation | A      | 0.845    | HIGH      | 0.867    | HIGH      |
| A3 ECM remodeling        | A      | 0.842    | HIGH      | 0.849    | HIGH      |
| A4 endothelium           | A      | 0.905    | HIGH      | 0.893    | HIGH      |
| A5 pericyte              | A      | 0.811    | HIGH      | 0.875    | HIGH      |
| A6 proteoglycan matrix   | A      | 0.887    | HIGH      | 0.861    | HIGH      |
| A7 basement membrane     | A      | 0.888    | HIGH      | 0.898    | HIGH      |
| A8 myofibroblastic CAF   | A      | 0.831    | HIGH      | 0.893    | HIGH      |
| A9 lymphatic endothelium | A      | 0.862    | HIGH      | 0.863    | HIGH      |

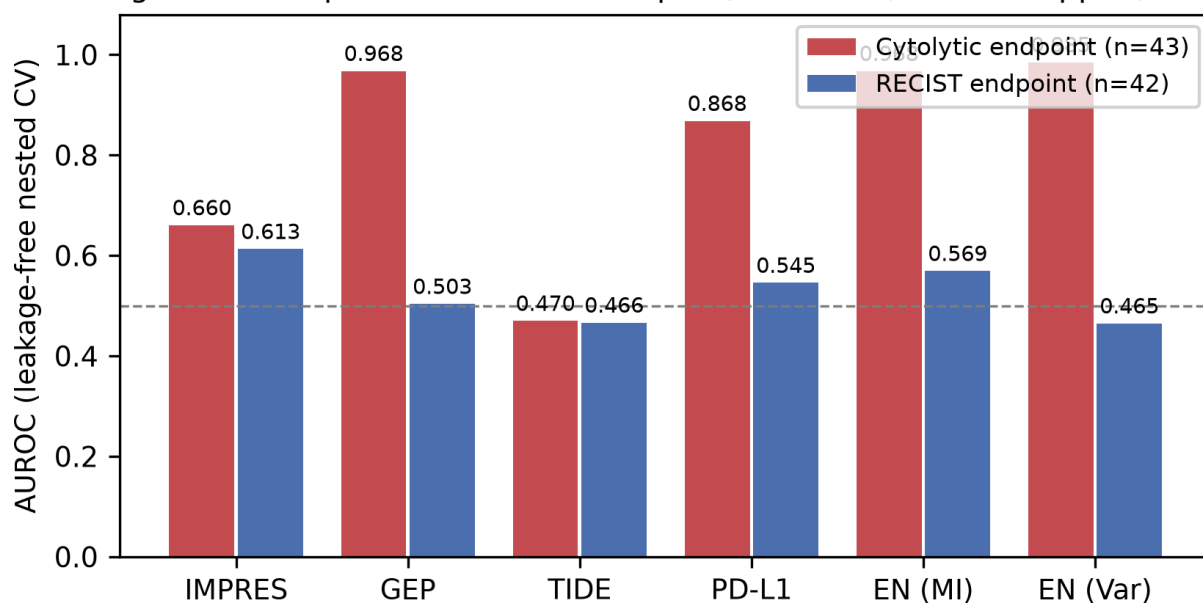
| Program                        | Family | Gide CDS | Gide risk | Hugo CDS | Hugo risk |
|--------------------------------|--------|----------|-----------|----------|-----------|
| A10 protease stroma            | A      | 0.830    | HIGH      | 0.883    | HIGH      |
| B1 cell cycle core             | B      | 0.823    | HIGH      | 0.869    | HIGH      |
| B2 E2F targets                 | B      | 0.906    | HIGH      | 0.883    | HIGH      |
| B3 MYC targets                 | B      | 0.811    | HIGH      | 0.821    | HIGH      |
| B4 DNA replication/repair      | B      | 0.861    | HIGH      | 0.870    | HIGH      |
| B5 spindle checkpoint          | B      | 0.878    | HIGH      | 0.882    | HIGH      |
| B6 translation machinery       | B      | 0.839    | HIGH      | 0.898    | HIGH      |
| B7 glycolysis                  | B      | 0.867    | HIGH      | 0.869    | HIGH      |
| B8 oxidative phosphorylation   | B      | 0.875    | HIGH      | 0.801    | HIGH      |
| B9 stress response             | B      | 0.840    | HIGH      | 0.852    | HIGH      |
| B10 unfolded protein response  | B      | 0.817    | HIGH      | 0.810    | HIGH      |
| C1 melanocytic differentiation | C      | 0.884    | HIGH      | 0.884    | HIGH      |
| C2 keratinocyte basal          | C      | 0.913    | HIGH      | 0.871    | HIGH      |
| C3 hypoxia                     | C      | 0.841    | HIGH      | 0.848    | HIGH      |
| C4 interferon response         | C      | 0.919    | HIGH      | 0.900    | HIGH      |
| C5 TNF/NF-kB response          | C      | 0.872    | HIGH      | 0.861    | HIGH      |
| C6 antigen presentation I      | C      | 0.839    | HIGH      | 0.870    | HIGH      |
| C7 complement                  | C      | 0.894    | HIGH      | 0.883    | HIGH      |
| C8 senescence                  | C      | 0.830    | HIGH      | 0.838    | HIGH      |
| C9 Wnt targets                 | C      | 0.792    | HIGH      | 0.806    | HIGH      |
| C10 MHC class II               | C      | 0.888    | HIGH      | 0.927    | HIGH      |

**Reading.** All 60 pseudo-endpoints — ECM/stromal, proliferation and lineage programs alike — score above the HIGH boundary, exactly like the cytolytic positive control, while random-label nulls stay lower (though with 3/10 replicates above 0.70 on the  $n = 28$  feature space). CDS therefore measures the gene-definedness of an endpoint, program-agnostically: it detects circular *construction*, cannot grade *content*, and its categorical thresholds carry no specificity among structured endpoints. This quantifies the C3-saturation limitation and motivates the CDS v2.0 redesign.

**Figures (v33; PNG files in `results/figures/v33/`, generated by `scripts/make_figures_v33.py`)**

**Figure S1. Response-definition collapse — bar plot (identical to main-text Figure 1)**

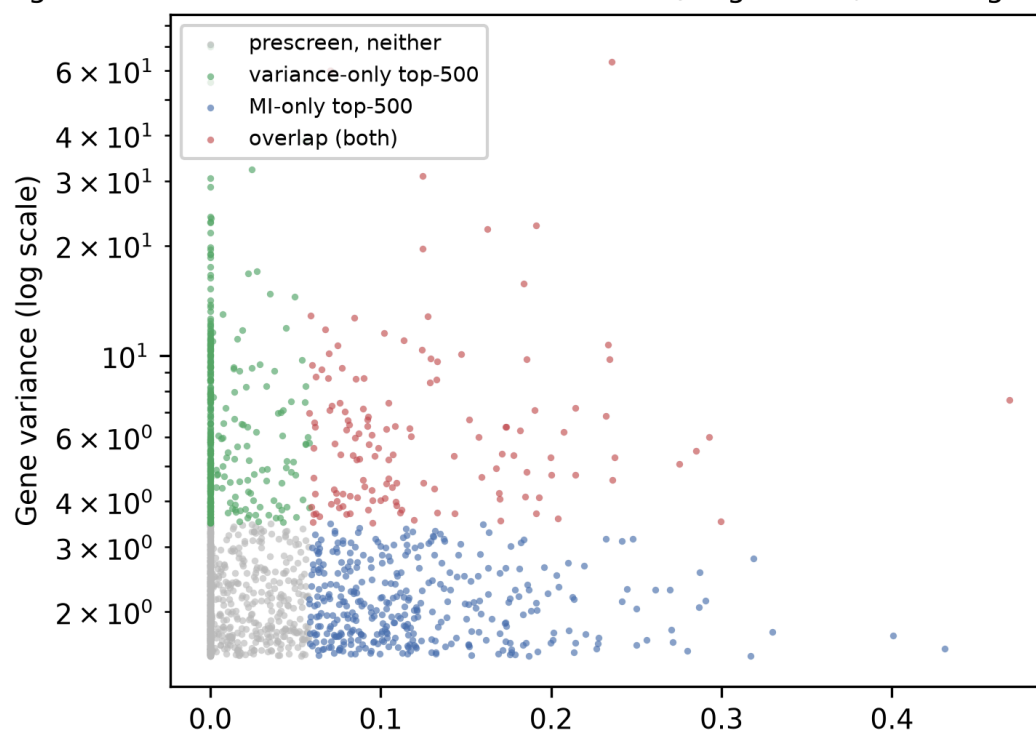
Figure S1. Response-definition collapse (Riaz 2017, HGNC-mapped, v33)



Riaz 2017, cytolytic vs RECIST AUROC per method (HGNC-mapped, v33). Same file as main-text  
Figure 1. figS1\_response\_definition\_collapse.png

## Figure S2. Feature selection — MI vs variance scatter

Figure S2. MI vs. variance feature selection (Hugo 2016, two-stage convention)

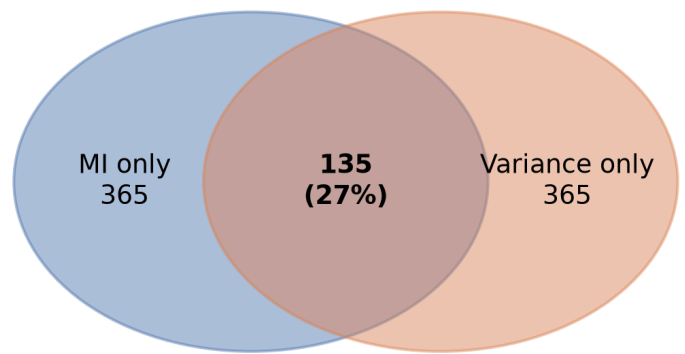


MI with response (stage-2, within top-2000 variance prescreen; full data n=28)

Gene-level MI (stage-2, within the top-2000 variance prescreen) vs variance, top-500 sets highlighted.  
figS2\_mi\_vs\_var.png

Figure S3. Gene-set overlap

Figure S3. Gene-set overlap: MI- vs. variance-selected

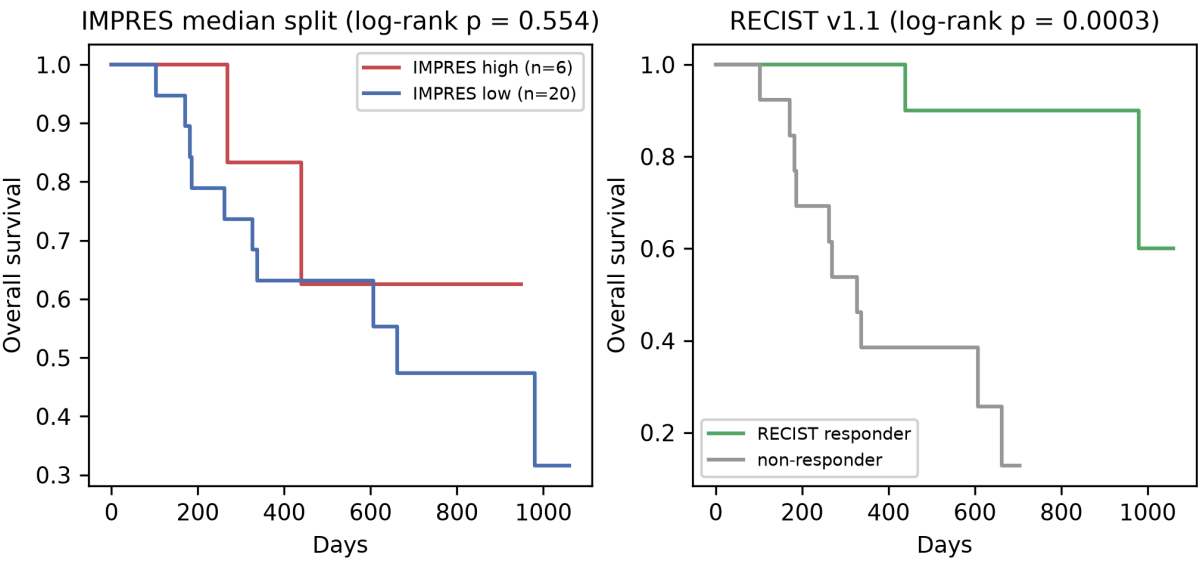


Top-500 genes, Hugo 2016 (n=28): stage-2 MI within top-2000 variance prescreen (v33 pipeline convention)

Top-500 overlap under the two-stage prescreen convention: 135/500 (27%). figS3\_gene\_overlap.png

Figure S4. Kaplan–Meier survival curves (Hugo 2016)

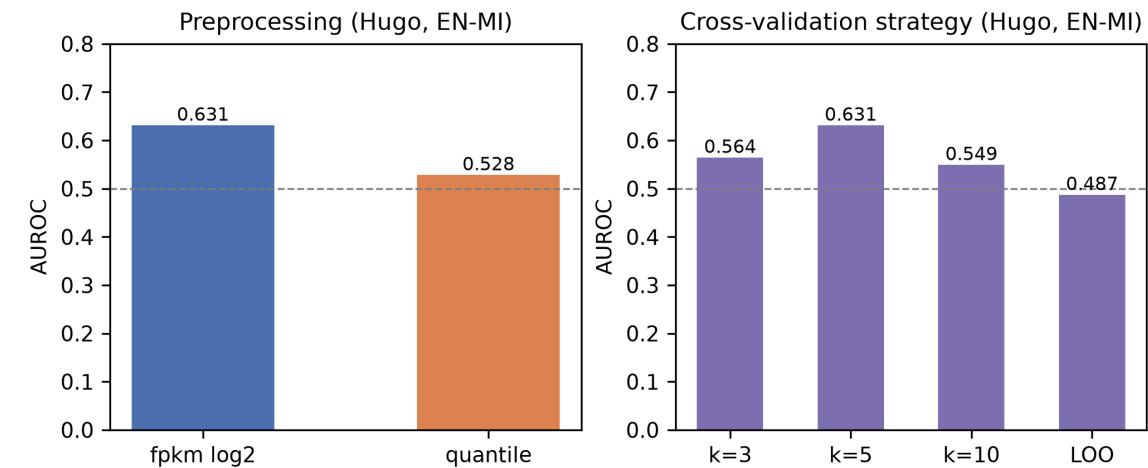
Figure S4. Hugo 2016 overall survival (true iAtlas OS\_STATUS events; n = 26, 12 events)



IMPRES median split and RECIST groups, v33 event convention (n = 25, 11 events). figS4\_km\_survival.png

Figure S5. Sensitivity analysis summary

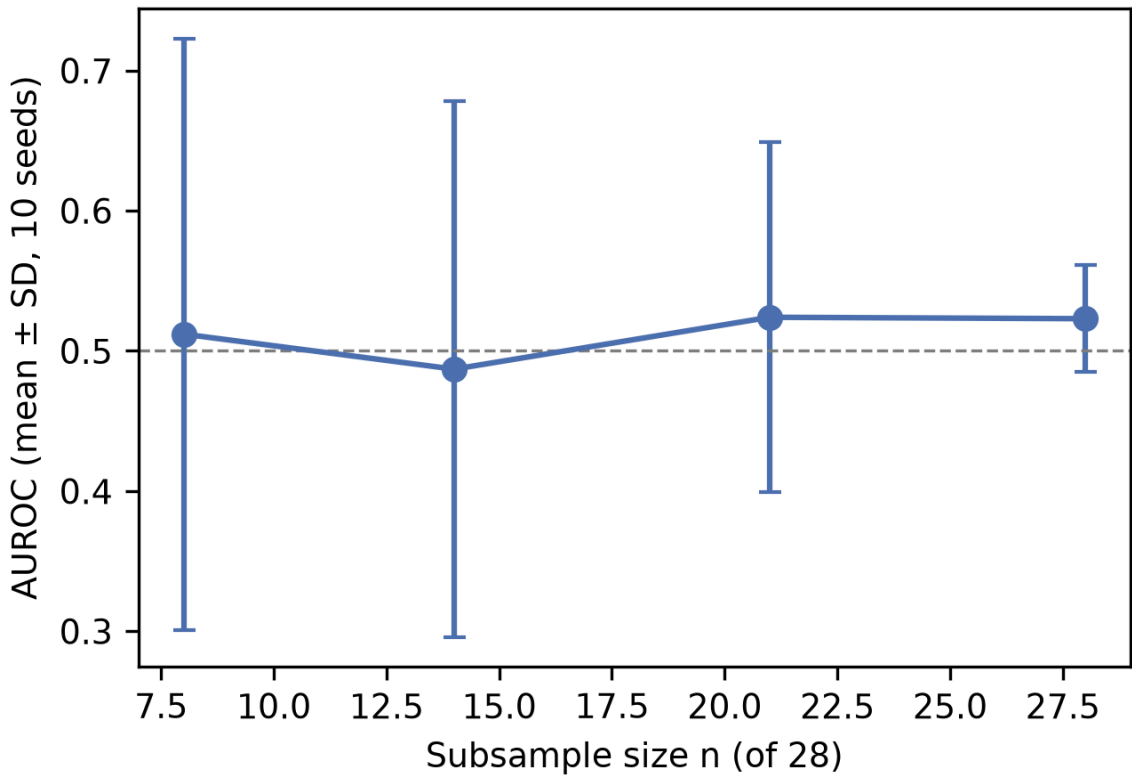
Figure S5. Sensitivity analyses (v33 protocol; TPM row not derivable from archived FPKM matrix)



Expression-scale and CV-strategy sensitivity (Hugo, EN-MI, v33). figS5\_sensitivity.png

Figure S6. Learning curves

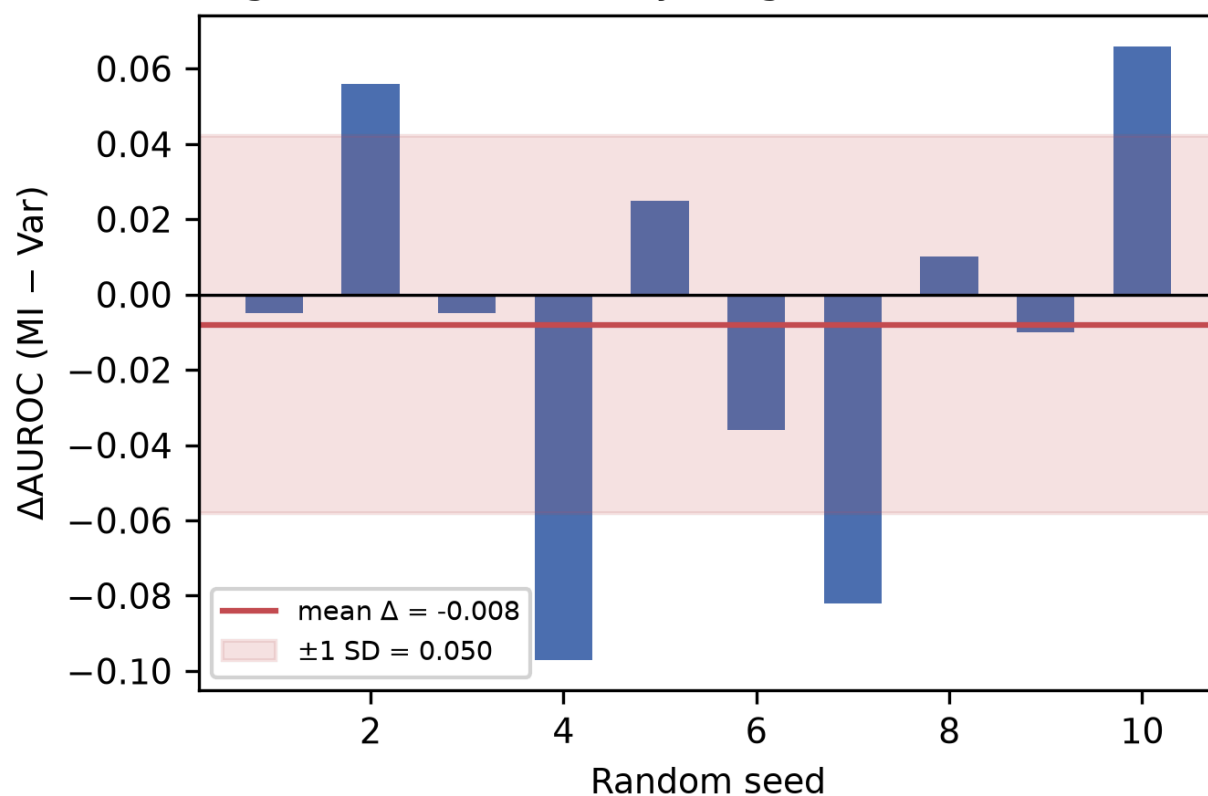
Figure S6. Learning curve (Hugo, ElasticNet-Var, v33 protocol)



ElasticNet-Var on Hugo subsamples, 10 seeds per size (v33). figS6\_learning\_curves.png

Figure S7. Random seed stability

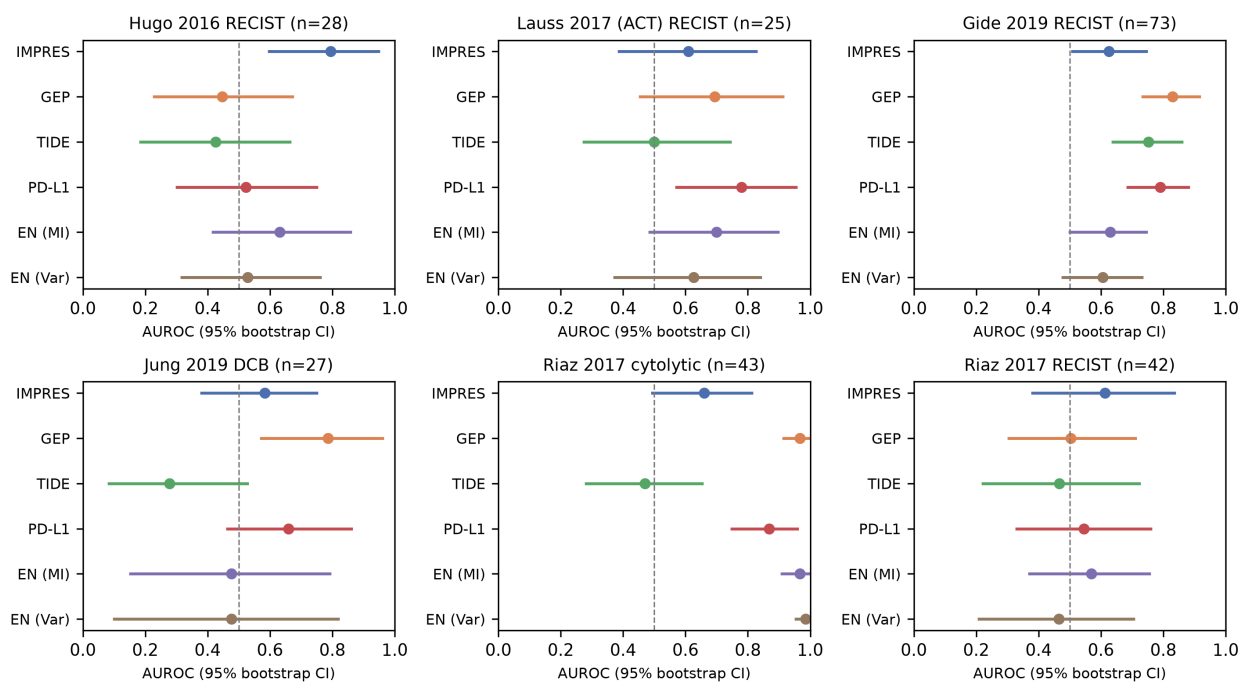
Figure S7. Seed stability (Hugo, 10 stratified seeds)



MI - Var  $\Delta\text{AUROC}$  across 10 stratified seeds (v33). figS7\_seed\_stability.png

Figure S8. Benchmark forest plot

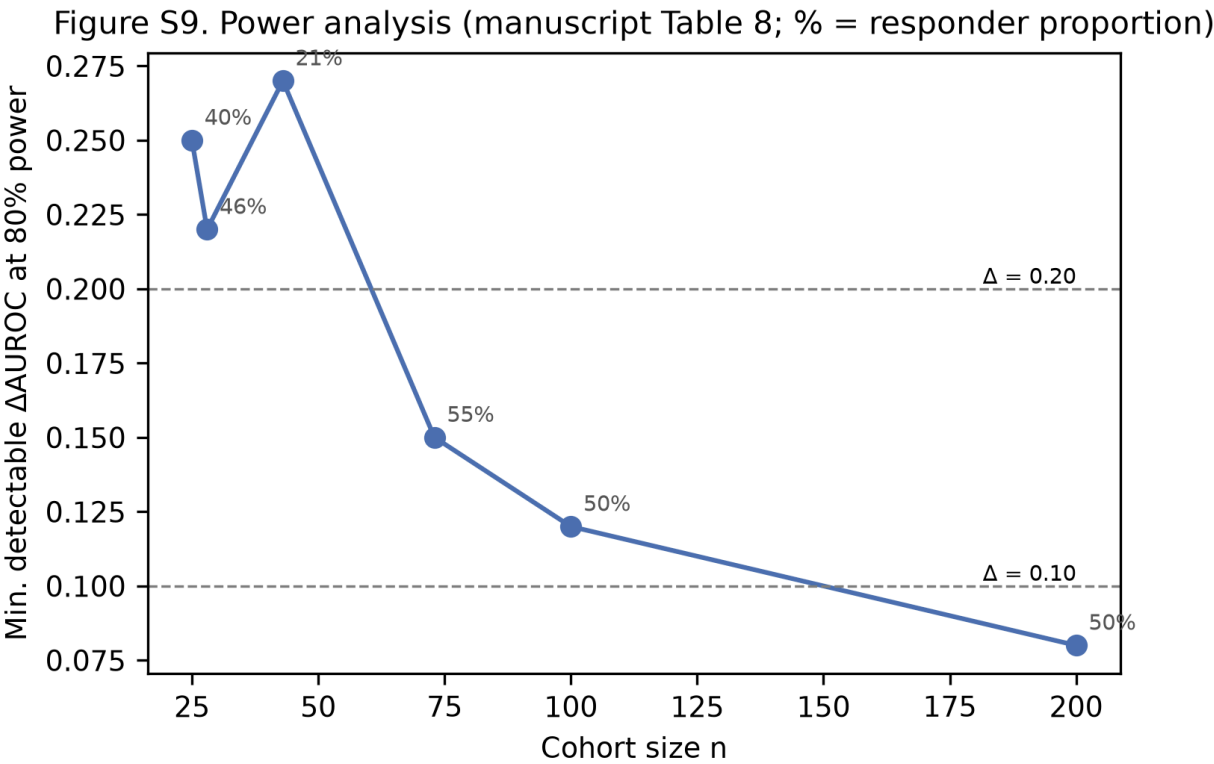
Figure S8. Benchmark forest plot — 6 cohort-endpoints  $\times$  6 methods (v33)



AUROC + 95% bootstrap CI, 6 cohort-endpoints  $\times$  6 methods. figS8\_forest.png

621

Figure S9. Power curves

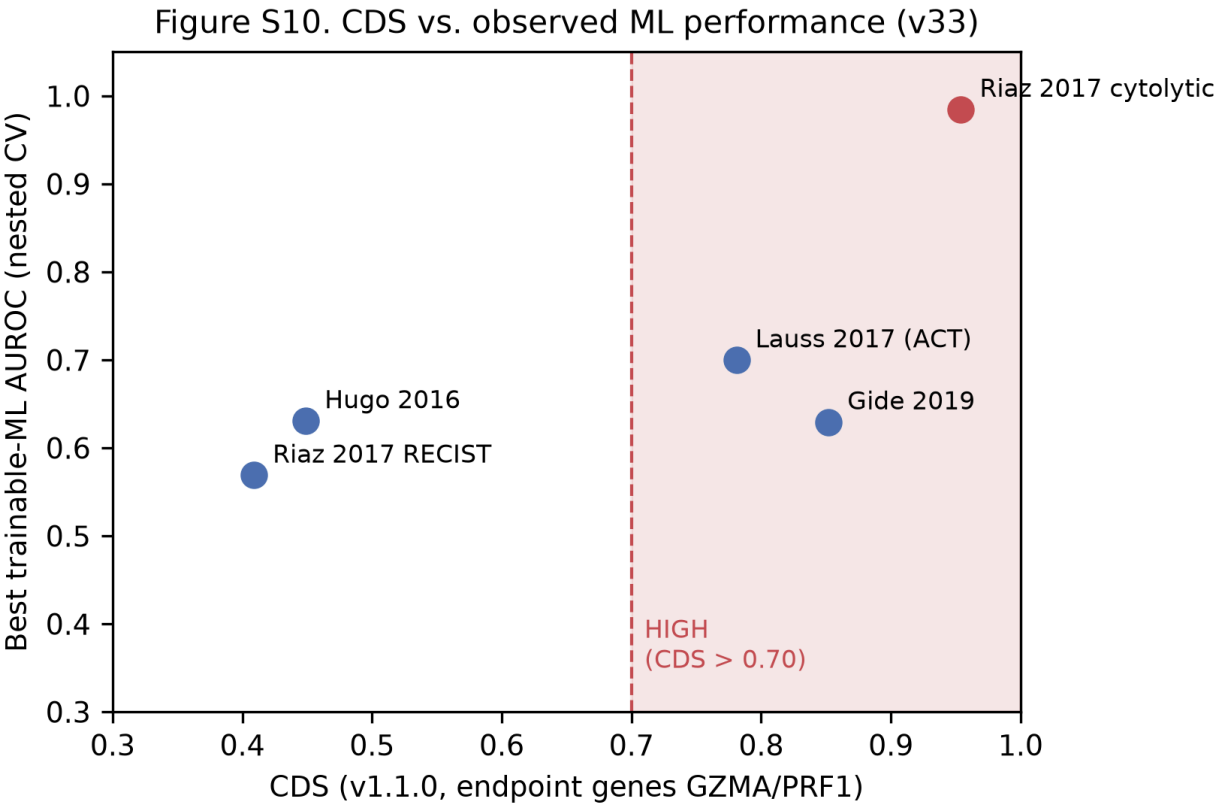


622

Minimum detectable  $\Delta$ AUROC at 80% power; points reproduce Table S25. figS9\_power\_curves.png

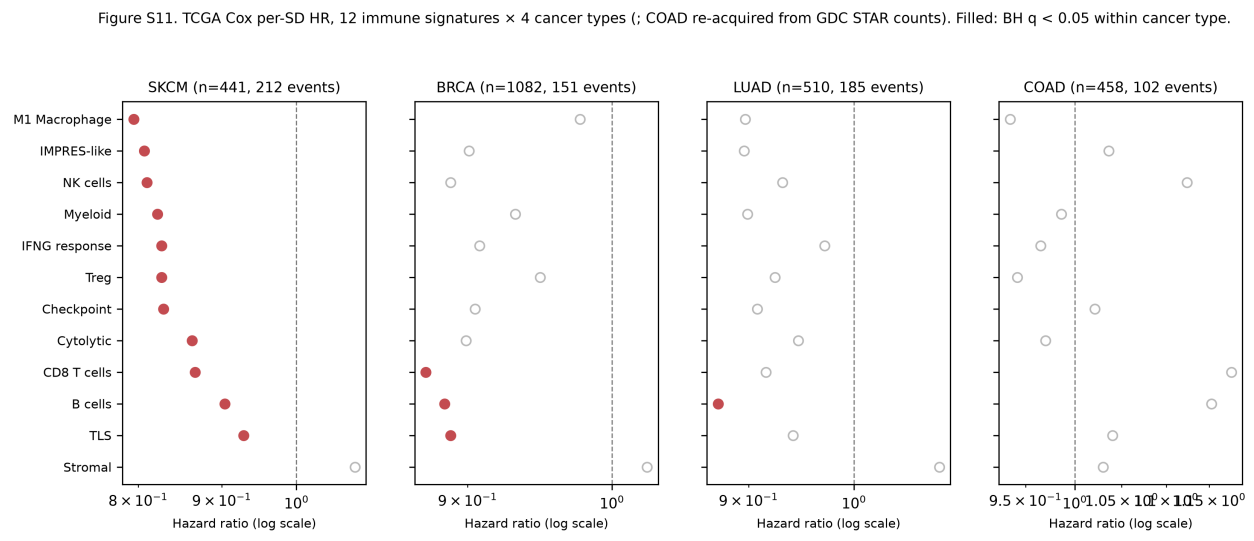
623

Figure S10. CDS vs observed ML performance (identical to main-text Figure 2)



CDS (v1.1.0 recomputation, Table S4b) vs best trainable-ML AUROC per endpoint; HIGH region shaded.  
Same file as main-text Figure 2. figS10\_cds\_vs\_auroc.png

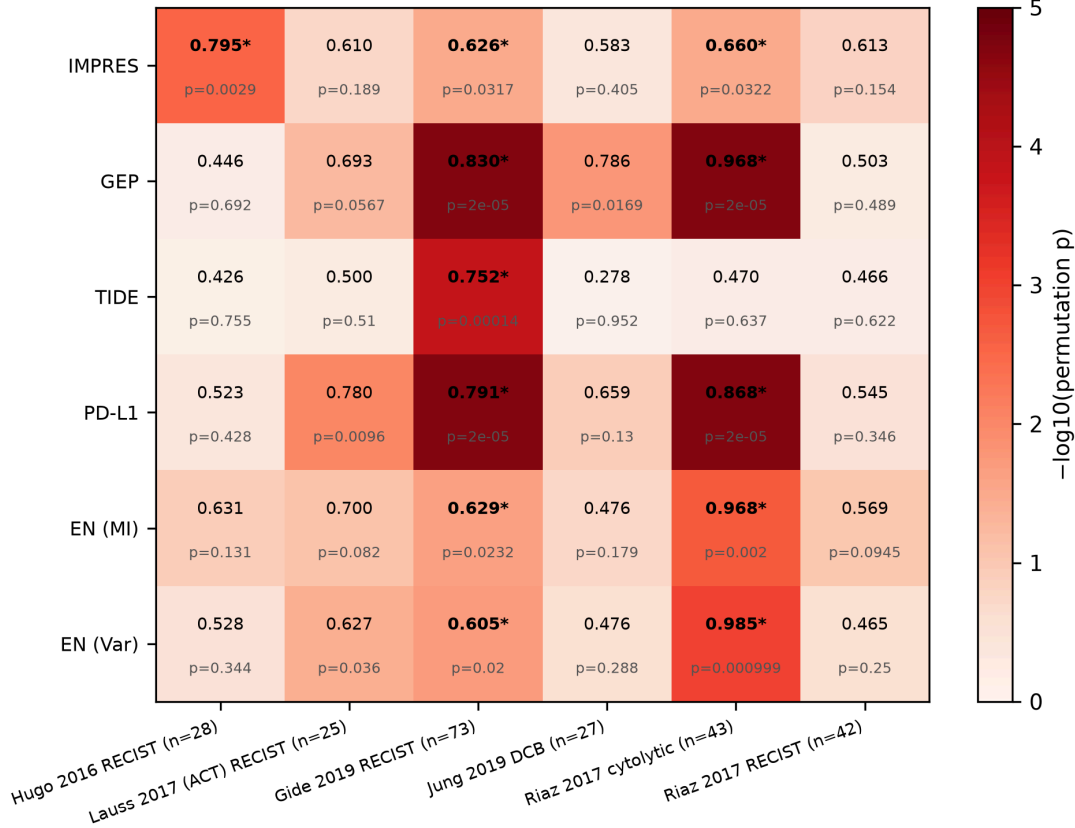
**Figure S11. TCGA forest plot**



Per-SD HR, 12 signatures  $\times$  4 cancer types (2,491 patients; COAD re-acquired from the GDC current index); filled = BH  $q < 0.05$  within cancer type. figS11\_tcga\_forest.png

**Figure S12. Complete permutation matrix (E4 heatmap; identical to main-text Figure 3)**

Figure S12. Complete permutation matrix (E4): observed AUROC + p per cell; \* BH  $q < 0.05$  within cohort. Fixed scorers: 50,000 prediction shuffles; ML: full-pipeline shuffles (per-cell n); n.p. = not permuted (PD-L1 excluded in v33 runs)



Observed AUROC + permutation p per cell; \* = BH  $q < 0.05$  within cohort. Same file as main-text Figure 3. figS12\_permutation\_heatmap.png

### Note S1. Synthetic Data Validation

On the bundled synthetic cohort (SYNTH\_TEST, n = 200; regenerated during the v33 validation fixes), the framework recovers the planted predictive signal: GEP AUROC = 0.679 on synthetic data with known ground-truth signal , IMPRES = 0.500 , ElasticNet-MI = 0.623 . The archived v28 note (n = 200, 2,000 genes; GEP 0.726 / EN-MI 0.546) refers to a superseded synthetic generator and is superseded by the current benchmark\_results\_VALIDATION.json.

### Note S2. TIDE Implementation Validation

v33 evaluates TIDE through the official tidepy v1.3.9 package directly. On Hugo 2016 RECIST, tidepy returns AUROC = 0.426; on Riaz 2017 (HGNC-mapped) TIDE is near chance on both endpoints (RECIST 0.466, cytolytic 0.470), consistent with the tidepy source check (classification = correlation['TIDE'] < vthres, default vthres = 0). The earlier simplified in-repo TIDE implementation (v28, 0.410 on Hugo) agreed with the official package when compared and has been retired in favor of the official package.

### Note S3. Cohort Acquisition and Quality Control (v33 updates)

---

Cohorts were downloaded from GEO (GEOquery) or cBioPortal (cBioPortalData; Gide 2019 via `mel_iatlas_gide_2019`). QC: pre-treatment status confirmation, removal of samples with missing response annotations, gene-level filtering, and gene-identifier format validation. v33 data-layer corrections, disclosed in the manuscript: (i) Riaz 2017 matrices stored `var_names` as Entrez Gene IDs — every symbol-based signature was degenerate (AUROC exactly 0.500); all Riaz results now come from the Entrez→HGNC-mapped matrices (21,769/22,187 genes mapped; `scripts/map_riaz_to_hgnc.py`); (ii) the Riaz RECIST h5ad was a transposed 42×43 label table and was rebuilt (`scripts/rebuild_riaz_recist.py`, `n = 42`, 22187 genes); (iii) Jung 2019 release codes 1 = NDB — the inverse of every other cohort; v33 uses 1 = DCB (6/27) after verification against GSE135222; (iv) the TCGA pipeline now validates TCGA barcode format and skips incompatible inputs with an explicit warning (the COAD withdrawal in Table S15).

### Note S4. TCGA Cox Regression — Full Methods (v33)

---

**Data.** TCGA RNA-seq and clinical survival data. SKCM/BRCA/LUAD use the cBioPortal pan-cancer-atlas RSEM matrices (SKCM: all tumour samples, `n = 441`, 212 deaths; BRCA `n = 1,082`; LUAD `n = 510`). COAD was re-acquired from the GDC current index (STAR counts, GENCODE v36, primary tumours, FPKM; clinical from the GDC demographic API; `scripts/build_coad_gdc.py`) after the original non-TCGA raw files forced the v33 withdrawal: `n = 458`, 102 deaths. For patients with multiple samples the first tumour sample per patient was retained. Non-numeric `OS_MONTHS` values are coerced to missing and dropped; `COAD OS_MONTHS = days / 30.436875` (GDC convention).

**Signatures.** 12 immune signatures (44 unique genes; the signature definitions are those of `scripts/run_tcga_cox.py`, overlapping but not identical to the 14-signature Table S10 set) as mean  $\log_2(\text{RSEM} + 1)$  of marker genes; the IMPRES-like signature uses the 12 IMPRES genes as mean expression (the pairwise-ratio logic is not applicable in the Cox pipeline).

**Statistics.** Univariate Cox models (lifelines `CoxPHFitter`); Wald p-values; **Benjamini–Hochberg correction within each cancer type (12 tests each)** — the archived analysis used nominal  $p < 0.05$  only and is superseded. Results: SKCM 11/12, BRCA 3/12, LUAD 1/12, COAD 0/12 signatures significant at  $\text{FDR} < 0.05$  (the restored COAD independently reproduces the archived 0/12 finding).

**Limitations.** Univariate models without clinical covariates; IMPRES-like signature is a mean-expression approximation; no time-dependent AUC is reported under v33 (the archived 0.33–0.60 range was not regenerated). Sample-type composition differs across the compared cancer types (SKCM metastatic-dominant; COAD primary-only); the SKCM decomposition in Table S15 shows the gradient signal is metastatic-carried and internally robust.

### Note S5. IMPRES Canonical Implementation — the 15 Pairwise Relations (v33)

---

v33 implements IMPRES exactly as the published algorithm: the score is the fraction of the following 15 logical relations (gene A > gene B) fulfilled by the sample, linearly rescaled when features are missing, with no retraining. The relations were decoded from the original authors' released feature set

([github.com/noamaus/IMPRES-codes](https://github.com/noamaus/IMPRES-codes), FEATS.mat) and cross-checked against the 9 pairs named in the paper text (9/9 match). The list below is extracted live from `workflows/methods/base_method.py` so it cannot drift from the code:

| #  | Relation         |
|----|------------------|
| 1  | CD274 > C10orf54 |
| 2  | CD86 > CD200     |
| 3  | CD40 > CD274     |
| 4  | CD28 > CD276     |
| 5  | CD40 > CD28      |
| 6  | TNFRSF14 > CD86  |
| 7  | CD27 > PDCD1     |
| 8  | CD28 > CD86      |
| 9  | CD40 > CD80      |
| 10 | CD40 > PDCD1     |
| 11 | CD80 > TNFSF9    |
| 12 | CD86 > HAVCR2    |
| 13 | CD86 > TNFSF4    |
| 14 | CTLA4 > TNFSF4   |
| 15 | PDCD1 > TNFSF4   |

**Table S19. External Replication — Response-Definition Effect on Liu 2019 (Independent Cohort, E1 executed)**

Pre-declared replication of Design Choice 1 on Liu 2019 (DFCI anti-PD-1 melanoma; n = 122 annotated samples, 48 RECIST responders; independent of all five benchmark cohorts; cBioPortal iAtlas harmonization, z-score expression — the benchmark metrics are invariant to per-gene affine transforms). Endpoints on identical patients: cytolytic surrogate (GZMA/PRF1 mean, median split — the Riaz construction) vs clinical RECIST (CR/PR). Leakage-free nested CV, frozen hyperparameters from `tune_primary`, seed 42. Machine-readable: `e1_liu2019_replication.json`; pipeline `scripts/e1_replication_liu2019.py`.

| Method          | Cytolytic AUROC | 95% CI        | RECIST AUROC | 95% CI        | $\Delta$ | Fixed-scorer perm p (cytolytic / RECIST) |
|-----------------|-----------------|---------------|--------------|---------------|----------|------------------------------------------|
| IMPRES          | 0.814           | [0.737-0.880] | 0.601        | [0.495-0.702] | +0.212   | 2e-05 / 0.0271 (n = 50000)               |
| GEP             | 0.959           | [0.928-0.986] | 0.535        | [0.427-0.642] | +0.425   | 2e-05 / 0.262 (n = 50000)                |
| TIDE            | 0.480           | [0.376-0.584] | 0.406        | [0.311-0.509] | +0.074   | 0.645 / 0.96 (n = 50000)                 |
| PD-L1 (CD274)   | 0.788           | [0.706-0.865] | 0.550        | [0.444-0.652] | +0.238   | 2e-05 / 0.18 (n = 50000)                 |
| ElasticNet (MI) | 0.956           | [0.917-0.984] | 0.538        | [0.423-0.637] | +0.418   |                                          |

| Method           | Cytolytic AUROC | 95% CI        | RECIST AUROC | 95% CI        | $\Delta$ | Fixed-scorer perm p (cytolytic / RECIST) |
|------------------|-----------------|---------------|--------------|---------------|----------|------------------------------------------|
| ElasticNet (Var) | 0.837           | [0.757-0.898] | 0.586        | [0.477-0.690] | +0.251   |                                          |

**CDS (declared genes GZMA/PRF1):** cytolytic surrogate 0.949 (HIGH; null mean 0.566) — the circular construction, as in Riaz; RECIST 0.721 (HIGH; null mean 0.591) — above its null mean, the genuine-biology signature also seen for Gide (Table S4b). Trainable-method bootstrap CIs on RECIST all include 0.5; the collapse therefore replicates without any trainable method reaching significance on the clinical endpoint.

Within-cohort BH over the four fixed scorers on the RECIST endpoint: IMPRES  $q = 0.108$ , GEP  $q = 0.350$ , TIDE  $q = 0.960$ , PD-L1  $q = 0.350$ . IMPRES is therefore nominally significant only ( $p = 0.027$ ,  $q = 0.108$ ) and does not survive the manuscript's own FDR framework.

*Provenance: GSE274975 (the originally pre-declared NSCLC candidate) was investigated over FTP (the only NCBI protocol reachable from the analysis machine) and disqualified: the series contains no RECIST, PFS or ICI-treatment annotations and cannot serve as an ICI replication cohort. Its complete pre-declared pipeline is retained (scripts/e1\_gse274975\_replication.py) for reuse on any suitable NSCLC ICI cohort. Liu 2019 is the disclosed substitution.*

## Table S20. Cross-Cancer Replication — Response-Definition Effect on IMvigor210 (Bladder Carcinoma, Non-Melanoma, E1 executed)

Pre-declared replication of Design Choice 1 on IMvigor210 (metastatic urothelial carcinoma, anti-PD-L1;  $n = 297$  response-annotated samples, 68 RECIST responders; independent of all five benchmark cohorts and of Liu 2019; cBioPortal iAtlas harmonization, z-score expression). This upgrades the replication from two-cohort to **cross-cancer**: the collapse replicates in melanoma (Riaz, Liu) and bladder carcinoma. Endpoints on identical patients: cytolytic surrogate (GZMA/PRF1 mean, median split — the Riaz construction) vs clinical RECIST (CR/PR). Leakage-free nested CV, frozen hyperparameters from tune\_primary, seed 42. Machine-readable: e1\_imvigor210\_replication.json; pipeline scripts/e1\_replication\_imvigor210.py.

| Method           | Cytolytic AUROC | 95% CI        | RECIST AUROC | 95% CI        | $\Delta$ | Fixed-scorer perm p (cytolytic / RECIST) |
|------------------|-----------------|---------------|--------------|---------------|----------|------------------------------------------|
| IMPRES           | 0.708           | [0.653-0.760] | 0.538        | [0.469-0.613] | +0.170   | 2e-05 / 0.161 ( $n = 50000$ )            |
| GEP              | 0.967           | [0.948-0.981] | 0.613        | [0.536-0.686] | +0.354   | 2e-05 / 0.00246 ( $n = 50000$ )          |
| TIDE             | 0.463           | [0.400-0.531] | 0.517        | [0.445-0.591] | -0.054   | 0.866 / 0.334 ( $n = 50000$ )            |
| PD-L1 (CD274)    | 0.863           | [0.819-0.902] | 0.566        | [0.493-0.638] | +0.297   | 2e-05 / 0.0519 ( $n = 50000$ )           |
| ElasticNet (MI)  | 0.951           | [0.927-0.970] | 0.657        | [0.587-0.731] | +0.294   |                                          |
| ElasticNet (Var) | 0.871           | [0.829-0.908] | 0.697        | [0.629-0.768] | +0.173   |                                          |

**CDS (declared genes GZMA/PRF1):** cytolytic surrogate 0.953 (HIGH; null mean 0.514) — the circular construction; RECIST 0.434 (MODERATE; null mean 0.492) — **below its null mean**, the cleanest CDS separation to date. Unlike Gide 2019 (CDS 0.852, above null max) and Liu 2019 (0.721, above null mean), IMvigor210's clinical endpoint shows no genuine-biology conflation with the declared genes — confirming that the prognostication-circularity conflation is cohort-specific.

**GEP on the clinical endpoint.** GEP is the only fixed scorer surviving FDR on the clinical RECIST endpoint ( $p = 0.0025$ , BH  $q = 0.010$  over the four fixed scorers). IMvigor210 is an application cohort of the GEP, which Ayers et al. (2017) developed on pembrolizumab KEYNOTE-series cohorts — the significance here reflects a weak effect (AUROC 0.613) detectable only at  $n = 297$ , consistent with the power analysis, not a cohort-of-origin effect.

**Trainable-method note.** At  $n = 297$ , ElasticNet-Var achieves AUROC = 0.697 with CI excluding 0.5 — the largest sample size at which a trainable method is detectable on a clinical endpoint in this study (Gide,  $n = 73$ , is the first such size by permutation FDR; Table S11), consistent with the power analysis.

## Table S21. Multiple-Comparison Family Sensitivity — BH $q$ under Three Family Definitions

All 36 permutation cells recomputed under: (A) within-cohort families, 6 x 6 cells (manuscript primary; conclusions are per-cohort-endpoint statements); (B) one global family of 36 cells; (C) families stratified by method class (fixed scorers 24 cells; trainable 12 cells). Generator: scripts/bh\_sensitivity\_v33.py -> bh\_sensitivity\_v33.json.

| Cohort               | Method         | p         | q (A within) | q (B global) | q (C stratified) | Sig A | Sig B | Sig C |
|----------------------|----------------|-----------|--------------|--------------|------------------|-------|-------|-------|
| Gide_2019_cBio       | PD_L1          | 2e-05     | 0.0001       | 0.0002       | 0.0001           | Yes   | Yes   | Yes   |
| Riaz_2017_cytolytic  | PD_L1          | 2e-05     | 0.0001       | 0.0002       | 0.0001           | Yes   | Yes   | Yes   |
| Gide_2019_cBio       | GEP            | 2e-05     | 0.0001       | 0.0002       | 0.0001           | Yes   | Yes   | Yes   |
| Riaz_2017_cytolytic  | GEP            | 2e-05     | 0.0001       | 0.0002       | 0.0001           | Yes   | Yes   | Yes   |
| Gide_2019_cBio       | TIDE           | 0.00014   | 0.0003       | 0.0010       | 0.0007           | Yes   | Yes   | Yes   |
| Riaz_2017_cytolytic  | ElasticNet_Var | 0.000999  | 0.0020       | 0.0060       | 0.0120           | Yes   | Yes   | Yes   |
| Riaz_2017_cytolytic  | ElasticNet     | 0.001996  | 0.0030       | 0.0103       | 0.0120           | Yes   | Yes   | Yes   |
| Hugo_2016            | IMPRES         | 0.0029    | 0.0174       | 0.0130       | 0.0116           | Yes   | Yes   | Yes   |
| Nathanson_2017       | PD_L1          | 0.0095998 | 0.0576       | 0.0384       | 0.0329           | -     | Yes   | Yes   |
| Jung_2019_DCB        | GEP            | 0.01692   | 0.1015       | 0.0609       | 0.0508           | -     | -     | -     |
| Gide_2019_cBio       | ElasticNet_Var | 0.02      | 0.0278       | 0.0655       | 0.0696           | Yes   | -     | -     |
| Gide_2019_cBio       | ElasticNet     | 0.0232    | 0.0278       | 0.0696       | 0.0696           | Yes   | -     | -     |
| Gide_2019_cBio       | IMPRES         | 0.03168   | 0.0317       | 0.0787       | 0.0774           | Yes   | -     | -     |
| Riaz_2017_cytolytic  | IMPRES         | 0.03224   | 0.0387       | 0.0787       | 0.0774           | Yes   | -     | -     |
| Nathanson_2017       | ElasticNet_Var | 0.03279   | 0.0716       | 0.0787       | 0.0716           | -     | -     | -     |
| Nathanson_2017       | ElasticNet     | 0.03579   | 0.0716       | 0.0805       | 0.0716           | -     | -     | -     |
| Nathanson_2017       | GEP            | 0.0567    | 0.0851       | 0.1201       | 0.1237           | -     | -     | -     |
| Riaz_2017_RECIST_v33 | ElasticNet     | 0.0976    | 0.4535       | 0.1952       | 0.1673           | -     | -     | -     |
| Jung_2019_DCB        | PD_L1          | 0.1299    | 0.3897       | 0.2461       | 0.2598           | -     | -     | -     |

| Cohort               | Method         | p       | q (A within) | q (B global) | q (C stratified) | Sig A | Sig B | Sig C |
|----------------------|----------------|---------|--------------|--------------|------------------|-------|-------|-------|
| Riaz_2017_RECIST_v33 | IMPRES         | 0.15352 | 0.4535       | 0.2763       | 0.2834           | -     | -     | -     |
| Nathanson_2017       | IMPRES         | 0.18926 | 0.2271       | 0.3244       | 0.3244           | -     | -     | -     |
| Riaz_2017_RECIST_v33 | ElasticNet_Var | 0.22673 | 0.4535       | 0.3700       | 0.3152           | -     | -     | -     |
| Jung_2019_DCB        | ElasticNet     | 0.23638 | 0.4595       | 0.3700       | 0.3152           | -     | -     | -     |
| Hugo_2016            | ElasticNet     | 0.28534 | 0.6420       | 0.4280       | 0.3342           | -     | -     | -     |
| Jung_2019_DCB        | ElasticNet_Var | 0.30635 | 0.4595       | 0.4411       | 0.3342           | -     | -     | -     |
| Riaz_2017_RECIST_v33 | PD_L1          | 0.34627 | 0.5194       | 0.4795       | 0.5540           | -     | -     | -     |
| Hugo_2016            | ElasticNet_Var | 0.36353 | 0.6420       | 0.4847       | 0.3635           | -     | -     | -     |
| Jung_2019_DCB        | IMPRES         | 0.40461 | 0.4855       | 0.5202       | 0.6042           | -     | -     | -     |
| Hugo_2016            | PD_L1          | 0.42797 | 0.6420       | 0.5313       | 0.6042           | -     | -     | -     |
| Riaz_2017_RECIST_v33 | GEP            | 0.48889 | 0.5867       | 0.5867       | 0.6446           | -     | -     | -     |
| Nathanson_2017       | TIDE           | 0.51029 | 0.5103       | 0.5926       | 0.6446           | -     | -     | -     |
| Riaz_2017_RECIST_v33 | TIDE           | 0.62231 | 0.6223       | 0.6944       | 0.7275           | -     | -     | -     |
| Riaz_2017_cytolytic  | TIDE           | 0.63655 | 0.6365       | 0.6944       | 0.7275           | -     | -     | -     |
| Hugo_2016            | GEP            | 0.69161 | 0.7552       | 0.7323       | 0.7545           | -     | -     | -     |
| Hugo_2016            | TIDE           | 0.75522 | 0.7552       | 0.7768       | 0.7881           | -     | -     | -     |
| Jung_2019_DCB        | TIDE           | 0.9521  | 0.9521       | 0.9521       | 0.9521           | -     | -     | -     |

**Reading.** The manuscript's primary conclusions are family-robust: (i) no trainable method survives FDR on any clinical endpoint at  $n \leq 43$  under any family definition (the minimum trainable  $q$  on an  $n \leq 43$  clinical endpoint is 0.08 under family B and 0.07 under family C — minima re-verified at extended permutation resolution after the 2026-09-18 trainable-cell extension); (ii) the circular-endpoint significance of ElasticNet/ElasticNet\_Var and GEP/PD-L1 on Riaz cytolytic holds under all three; (iii) IMPRES on Hugo and GEP/TIDE/PD-L1 on Gide hold under all three. Four cells are family-dependent and are reported as within-cohort results throughout the manuscript: ElasticNet\_Var on Gide ( $q$  0.028/0.066/0.070), IMPRES on Gide (0.032/0.083/0.077), IMPRES on Riaz cytolytic (0.039/0.083/0.077), and PD-L1 on Lauss (0.058/0.038/0.033 — this cell becomes FDR-significant under the wider families; the manuscript's nominal-only wording is the conservative reading).

**Table S22. AUROC across Four Clinical-Endpoint Cohorts (v33, HGNC-mapped; 95% CIs from 1,000 bootstrap iterations where shown; migrated from the main text)**

| Method           | Hugo 2016 (28)             | Lauss 2017 (25) | Gide 2019 (73) | Riaz RECIST (42)    |
|------------------|----------------------------|-----------------|----------------|---------------------|
| IMPRES           | <b>0.795</b> [0.597–0.947] | 0.610           | 0.626          | <b>0.613</b>        |
| GEP              | 0.446                      | 0.693           | <b>0.830</b>   | 0.503               |
| TIDE             | 0.426                      | 0.500           | 0.752          | 0.466               |
| PD-L1 (CD274)    | 0.523                      | <b>0.780</b>    | 0.791          | 0.545               |
| ElasticNet (Var) | 0.528 [0.318–0.761]        | 0.627           | 0.605          | 0.465 [0.208–0.704] |
| ElasticNet (MI)  | 0.631 [0.417–0.857]        | 0.700           | 0.629          | 0.569 [0.370–0.755] |

† Permutation status and BH q-values in Table S11. Riaz RECIST n = 42 (rebuilt cohort). Bold = best AUROC per column for context only; no method ranking is claimed (all bootstrap CIs of the trainable methods on clinical endpoints overlap 0.5 at  $n \leq 43$ ).

**Table S23. DCB Endpoint, Jung 2019 (n = 27, NSCLC; corrected labels: 1 = durable clinical benefit, 6/27; GSE135222)**

| Method           | AUROC | 95% CI      |
|------------------|-------|-------------|
| IMPRES           | 0.583 | 0.380–0.750 |
| GEP              | 0.786 | 0.573–0.960 |
| TIDE             | 0.278 | 0.083–0.527 |
| PD-L1 (CD274)    | 0.659 | 0.464–0.861 |
| ElasticNet (MI)  | 0.476 | 0.152–0.792 |
| ElasticNet (Var) | 0.476 | 0.100–0.818 |

TIDE scores far below chance on this cohort (0.278); with only 6 DCB responders this deviation sits inside the permutation null envelope ( $p = 0.952$ ), and we report the value as computed under the tidepy default threshold convention rather than re-tuning it to the cohort.

**Table S24. Overall Survival Stratification on Hugo 2016 (true events)**

| Cohort                  | n  | Events | Median OS | IMPRES KM p | RECIST KM p   | GEP KM p |
|-------------------------|----|--------|-----------|-------------|---------------|----------|
| Hugo 2016 (true events) | 26 | 12     | 439 days  | 0.554       | <b>0.0003</b> | 0.475    |

Hugo values come from true OS events (iAtlas OS\_STATUS, 26 patients / 12 deaths; matching validated by 24/27 local OS\_days agree with iAtlas OS\_MONTHS ( $\pm 60$  d); results/benchmark/v33/hugo\_survival\_true.json). The local GSE100797 (Lauss) matrix contains no OS metadata; the archived Lauss survival row from earlier versions could not be regenerated under the v33 protocol and is therefore excluded from this table (archived values retained in Table S9 for continuity). RECIST response strongly stratifies overall survival on Hugo (log-rank  $p = 0.0003$ ). IMPRES — the best AUROC method on Hugo 2016 — does not significantly stratify OS (KM  $p = 0.554$  with true events). Cox regression with z-standardized signatures (HR per 1 SD) confirms: IMPRES HR = 0.631 (95% CI [0.325–1.227],  $p = 0.175$ ) on Hugo, GEP HR = 1.03 (95% CI [0.562–1.887],  $p = 0.923$ ). AUROC for binary response prediction does not guarantee clinically meaningful survival stratification. Clinical utility metrics for ElasticNet\_Var on Hugo 2016 (v33 recomputed; Table S9): **DCA** net benefit 0.345 vs treat-all 0.405 at  $p_t = 0.10$  and 0.259 vs 0.330 at 0.20 — positive but consistently below "treat all"; at 0.50 the model is at 0. **NRI vs. random baseline** =  $-0.113$  ( $p = 0.658$ , not significant). **IDI** =  $-0.029$  (sensitivity 0.538, specificity 0.533).

**Table S25. Minimum Detectable  $\Delta$ AUROC at 80% Power ( $\alpha = 0.05$ , two-sided; Hanley–McNeil paired model,  $r = 0.75$ , the most favorable assumption; exact values from scripts/power\_table\_v33.py)**

| n   | Responder % | Min Detectable $\Delta$ AUROC | Example Cohort   |
|-----|-------------|-------------------------------|------------------|
| 25  | 40%         | 0.23                          | Lauss 2017 (ACT) |
| 28  | 46%         | 0.21                          | Hugo 2016        |
| 42  | 21%         | 0.21                          | Riaz RECIST      |
| 73  | 55%         | 0.13                          | Gide 2019        |
| 100 | 50%         | 0.11                          | —                |
| 200 | 50%         | 0.08                          | —                |

At  $n = 25$ – $43$ , the minimum detectable  $\Delta$ AUROC is  $0.21$ – $0.23$ . For reliable pairwise comparison at  $\Delta$ AUROC =  $0.10$  (80% power), the required sample size is  $\approx 130$  under the most favorable correlation assumption (paired,  $r = 0.75$ ) and  $\approx 260$ – $510$  under weaker or no correlation (power\_table\_v33.json);  $n \geq 50$  is therefore reported as a minimum practical floor, not as a sufficient condition for adequately powered pairwise comparison.

## Note S6. Negative-Control Stress Test — Design and Interpretation (E2)

**Motivation.** The internal CDS validation used four real endpoints and a synthetic null; the question left open was whether CDS separates 'gene-defined circular construction' from 'any structured biology'. This stress test answers it.

**Design.** Thirty canonical gene programs per cohort in three families: (A) ECM/stroma without EMT-TFs (10 sets: fibrillar collagens, fibroblast activation, ECM remodeling, endothelium, pericyte, proteoglycan matrix, basement membrane, myofibroblastic CAF, lymphatic endothelium, protease stroma); (B) tumour purity/proliferation (10 sets: cell cycle, E2F, MYC targets, DNA replication/repair, spindle checkpoint, translation, glycolysis, OXPHOS, stress response, UPR); (C) lineage/signalling programs (10 sets: melanocytic differentiation, keratinocyte basal, hypoxia, interferon, TNF/NF- $\kappa$ B, antigen presentation, complement, senescence, Wnt targets, MHC-II). Each pseudo-endpoint is the median split of the program-mean expression — the identical construction to the cytolytic surrogate — with the program's own genes declared to CDS. Cohorts: Gide 2019 ( $n = 73$ ) and Hugo 2016 ( $n = 28$ ). References: the cytolytic surrogate on the same cohort and a 10-replicate random-label null. Gene lists are in scripts/cds\_negative\_controls.py; per-endpoint CDS/C1/C2/C3 in cds\_negative\_controls.json.

**Result.** 60/60 pseudo-endpoints score HIGH (Gide range  $0.792$ – $0.942$ , mean  $0.861$ ; Hugo  $0.801$ – $0.946$ , mean  $0.869$ ), indistinguishable in band from the cytolytic positive control ( $0.955 / 0.923$ ), while the random-label nulls sit lower (Gide mean  $0.518$ , max  $0.726$ ; Hugo mean  $0.608$ , max  $0.764$  — note 3/10 Hugo replicates still exceed the shipped  $0.70$  boundary at  $n = 28$ ).

**Interpretation.** Both readings are reported and both are used in the manuscript. (1) *Program-agnostic sensitivity*: CDS correctly flags every operationally gene-defined surrogate, whatever the program — the construction itself, not the biology, is what triggers the score; a median-split gene-program endpoint IS circular by construction, so HIGH is the correct answer for all 60. (2) *No specificity among structured endpoints*: because real programs and the circular surrogate land in the same band, and because random labels on small real cohorts can also cross the boundary, the categorical HIGH label cannot distinguish circularity from genuine gene-label biology (the Gide 2019 RECIST case, Table S4b). The validated use of CDS v1.1.0 is therefore the continuous value read against its cohort-matched null, plus the procedural

rule that declaring endpoint-defining genes identifies circular construction. Both motivate the permutation-based C3 in CDS v2.0.

**Table S26. Clinical–Clinical Response-Definition Contrast on Liu 2019 — RECIST vs PFS-derived DCB (first non-circular contrast, E5)**

One public cohort carries two clinical response definitions on the same patients: the iAtlas harmonization of Liu 2019 annotates both RECIST (CR/PR; tumour-biopsy rows) and PFS (patient-level rows). DCB = progression-free at  $\geq 6$  months; samples censored before 6 months are excluded (event-free status unknown). Expression: the iAtlas z-score matrix (33,732 of 59,409 genes retained after dropping genes unmeasured in any sample). Leakage-free nested CV with frozen hyperparameters from tune\_primary, seed 42; fixed-scorer permutations 50,000 shuffles; CDS v1.1.0 with declared genes GZMA/PRF1 and 10-replicate random-label nulls. Pipeline: scripts/e5\_clinical\_contrast\_liu2019.py; machine-readable: e5\_liu2019\_clinical\_contrast.json. RECIST-column AUROCs differ from Table S19 by  $\leq 0.013$  for the four fixed scorers; the two trainable methods deviate more (ElasticNet-Var 0.441 vs 0.586, ElasticNet-MI 0.554 vs 0.538) because their per-fold feature selection is re-fit on the complete-case-filtered matrix (33,732 of 59,409 genes). All conclusions are within-experiment comparisons, so this drift does not affect the RECIST-vs-DCB contrast.

Endpoint concordance:  $\kappa(\text{RECIST}, \text{DCB}) = 0.669$ ; 48 RECIST responders, 58 DCB; cross-tab (rows RECIST 0/1  $\times$  cols DCB 0/1): 59/15/5/43. Samples progressing before 6 months are scored NDB; samples censored before 6 months would be excluded (none in this cohort); samples censored at or after 6 months count as DCB (event-free at the landmark). IMPRES permutation p-values in this table are nominal, not BH-corrected within the E5 design.

| Method | RECIST AUROC        | 95% CI              | DCB AUROC | 95% CI          | $\Delta$ (RECIST–DCB) | Fixed-scorer perm p (RECIST / DCB) |
|--------|---------------------|---------------------|-----------|-----------------|-----------------------|------------------------------------|
| IMPRES | 0.614 [0.515–0.714] | 0.633 [0.532–0.729] | -0.019    | 0.0163 / 0.0053 |                       |                                    |

| GEP | 0.532 [0.426–0.641] | 0.554 [0.453–0.660] | -0.021 | 0.2763 / 0.1516 |  
| TIDE | 0.398 [0.302–0.502] | 0.446 [0.344–0.545] | -0.048 | 0.9709 / 0.8447 |  
| PD-L1 (CD274) | 0.550 [0.444–0.652] | 0.601 [0.503–0.700] | -0.051 | 0.1795 / 0.02606 |  
| ElasticNet (MI) | 0.553 [0.450–0.660] | 0.554 [0.452–0.662] | -0.000 | |  
| ElasticNet (Var) | 0.441 [0.334–0.553] | 0.524 [0.419–0.634] | -0.083 | |

**CDS** (declared genes GZMA/PRF1): RECIST 0.721 (null mean 0.622); DCB 0.726 (null mean 0.549) — both in the genuine-biology band, far below the circular surrogate's 0.954.

**Reading.** The largest endpoint-switch delta across six methods is 0.084 (ElasticNet-Var) — roughly sixfold below the circular collapse (0.520 on Riaz) and below the same-endpoint between-method gap on Liu RECIST (0.216, IMPRES vs TIDE). Two clinical definitions of the same construct barely move predictions: the response-definition effect demonstrated in Design Choice 1 is attributable to the circularity channel (gene-defined surrogates), not to clinical-definition disagreement — closing the

968

969

boundary acknowledged in earlier versions and confirming the mechanistic prediction of the Discussion's predictive account.